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AD2428 A2B Transceiver Audio Performance Testing and Results

Contributed by Tomarakos, John *Rev 1 – September 26, 2024*

Introduction

Systems based on the AD242x A^{2}B™ (Automotive Audio Bus) transceiver deliver multi-channel, bi-directional I^{2}S/TDM audio, clock, and data over a maximum distance of 80 meters using unshielded twisted-pair cable. The AD242x family includes AD2426, AD2427, and AD2428 devices. The audio performance of such systems, which perform audio clock recovery from the transmitted stream, can be subject to some degradation due to clock jitter. Clock jitter accumulates as the clock information proceeds from node to node over the A^{2}B bus. This increase in jitter can result in a decrease in audio quality for audio devices sensitive to clock jitter. For most engineers designing automotive-grade audio systems, this degradation in audio signal quality may not be sufficient to cause concern or may be acceptable. For higher-end or premium automotive systems with more stringent audio quality requirements, a closer look at audio performance is warranted.

This application note:

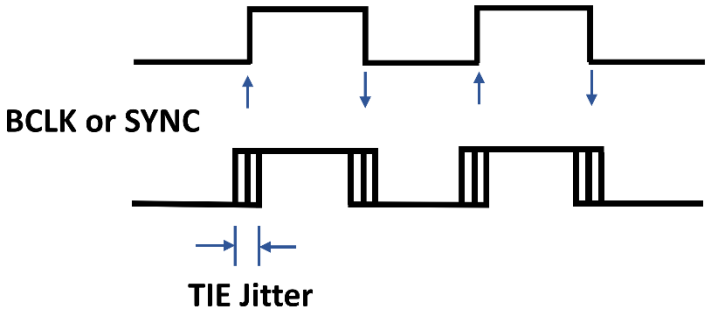
- Evaluates the audio performance of an AD242x A^{2}B bus.
- Describes results of testing performed on an AD2428 A^{2}B bus.
- Discusses possible methods to mitigate audio degradation for A^{2}B-connected systems.

Jitter Accumulation in A^{2}B Systems

A^{2}B networks use a daisy-chain style topology, where a clock signal propagates and is recovered at each node. The A^{2}B transceiver at the main node uses the SYNC and BCLK signals received at the I^{2}S/TDM interface for the master clock of the entire A^{2}B bus. This main node generates clocks and encodes them into the bus data from which each downstream sub node derives its own synchronization signals. Each subordinate node subsequently regenerates the clocks and embeds it into the A^{2}B data for the next downstream node. The clock that is propagated along the cascaded nodes may show increased jitter as the number of nodes increases. While the data on the bus is bit-correct at all points, each node contributes a small amount of jitter to the clocks resulting in a slight increase in overall jitter as the node count increases. There are several ways to measure this jitter (Figure 1). The most common methods for clock recovery systems are cycle-to-cycle jitter and Time Interval Jitter (TIE).

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Figure 1 Time Interval (TIE) Jitter



Time Interval Error (TIE) is the difference between the observed clock edge time and expected clock edge time. *TIE* refers to clock edges, while *period jitter* refers to the clock period. TIE is calculated by subtracting the actual clock edge from the ideal clock edge. Analog Devices provides RMS Time Interval Error (TIE_{RMS}) data measured at each subordinate node I^2S/TDM port in the *AD2426/AD2427/AD2428 A2B Transceiver Datasheet*[1]. Tables 10 of the AD242x datasheet (also displayed below in Figure 2) provides each node's RMS TIE jitter for the I^2S SYNC signal up to a maximum of 16 supported subordinate nodes. The accumulated jitter shown in the table does not result in bit errors. However, it can be observed at each sub node I^2S/TDM port and can result in distortion during any A/D and D/A conversions.

Figure 2 AD2428 Datasheet Specified TIE_{RMS} Jitter for I^2S SYNC Signal

Table 10. SYNC Output RMS TIE Jitter at Each Subordinate Node[1]

Subordinate Node	Typ	Max	Unit
1	1.57		ns
2	1.79		ns
3	1.91		ns
4	2.04		ns
5	2.15		ns
6	2.27		ns
7	2.44		ns
8	2.47		ns
9	2.58		ns
10	2.70	5.50	ns
11	2.85	5.66	ns
12	2.97	5.82	ns
13	3.09	5.98	ns
14	3.21	6.14	ns
15	3.33	6.30	ns
16	3.45	6.46	ns

[1] The jitter specification has spread-spectrum clocking disabled.

Prior to measuring THD+N (Total Harmonic Distortion and Noise – see the next section in this application note for more details), clock jitter measurements were first taken using the Audio Precision APx555

connecting to the I²S/TDM port of each AD2428 sub node. These results are displayed in Table 1. Notice that these measurements were slightly higher than the datasheet typical values shown in Table 10 of the datasheet. The parts used in this experiment were selected because they have higher jitter at 25 deg C than the typical A2B part. Despite the increase in TIE_{RMS} with increasing node count, no degradation of THD+N was observed through higher node counts when analyzing the digital serial data measured from I²S/TDM input at the main node to I²S/TDM output at each sub node.

Table 1 Measured SYNC clock Jitter for 16 Subordinate Nodes Using the APx555 Audio Analyzer

Sub Node	Typical TIE_{RMS} (ns)	Sub Node (cont.)	Typical TIE_{RMS} (ns)
0	1.82	8	3.25
1	2.14	9	3.41
2	2.35	10	3.58
3	2.64	11	3.70
4	2.79	12	3.82
5	2.94	13	3.94
6	3.05	14	4.05
7	3.13	15	4.19

Audio Performance Measurement-Total Harmonic Distortion Plus Noise

The primary goal for designing high-quality audio systems is to ensure that the device output can faithfully transmit or reconstruct the audio signal that is fed into the system. And that means ideally, the output signal is an identical or scaled version of the input signal. Anything that alters the input signal in any way, such high-jitter clock on the D/A conversion stage, is known as distortion. Audio performance and quality (SNR, dynamic range, THD, and so forth) must be measured and quantified.

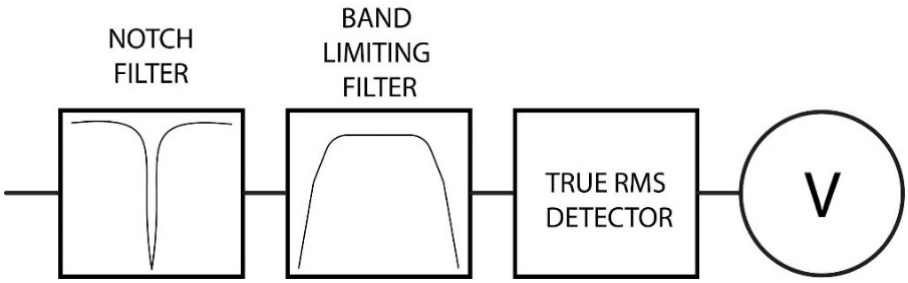
The best way to test the audio quality and performance of such a system is to simulate or inject into the system a pure sine tone and perform spectral analysis of the output signal. This helps us analyze the output of the system for any unwanted distortion components that were not part of the original signal input. These nonlinearities cause harmonic distortion at multiples of the input sine tone frequency generated and other noise components.

The most common method of measuring distortion and noise components (Figure 3) is THD+N, done by:

1. Implementing a notch filter tuned to the generated or measured fundamental sine tone frequency.
2. Following the notch filter with a band limiting filter (normally set within our audible hearing range of 20 Hz to 20 kHz).
3. Ending with an RMS level detector that determines the ratio of the residual noise and harmonic divided by the level of the input signal.

For the A²B system, use the Audio Processor (AP) to inject a tone on the I²S input and measure the THD+N at the far-end DAC analog output using the AP. To perform this test, Analog Devices used the APx555 Audio Precision Analyzer and APX500 version 8.0 software.

Figure 3 Block Diagram of THD+N Analyzer (Source: ap.com)

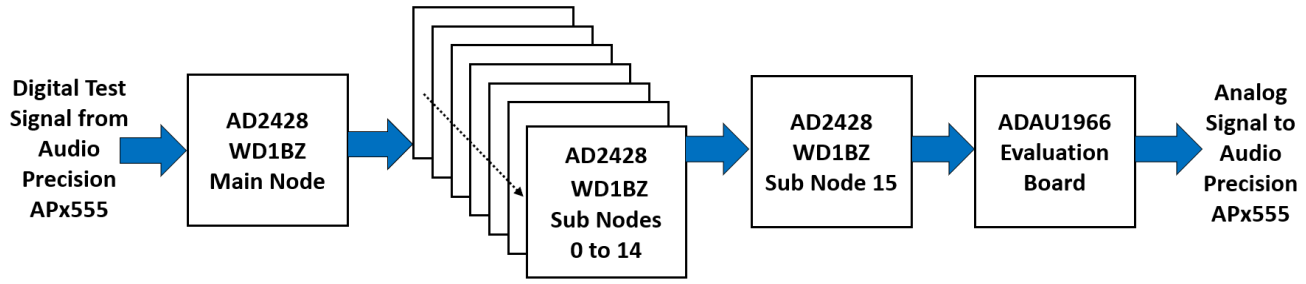


The AD2428 transceiver uses a built-in phase-locked loop (PLL) to recover the clock at each node. A small amount of jitter measured in single-digit nanoseconds happens at each node, resulting in a small amount of audio degradation that is additive as distance from the main node increases. The audio degradation is not tonal in nature but spread across the spectrum. With high-level signals, this results in earlier sub-nodes in an AD2428 transceiver-based system having slightly better measured audio performance than sub-nodes towards the end of the A^{2}B bus. The measured difference between nodes cannot be observed with lower-level audio signals; therefore, there is no audible decrease in audio performance with higher node counts and lower signal levels.

THD+N Measurement Setup and Equipment Required

The audio performance of the AD242x A^{2}B bus can be measured using a series of AD242x transceiver evaluation boards, an audio performance measurement system, and a digital-to-analog converter. This test used the Audio Precision APx555 Audio Analyzer with the Analog Devices ADAU1966 DAC as shown in the Figure 4 block diagram. See the *ADAU1966 16-Channel High Performance Differential Output, 192 kHz, 24-Bit DAC Datasheet*[8] for further setup information for this test.

Figure 4 Block Diagram of Test Setup

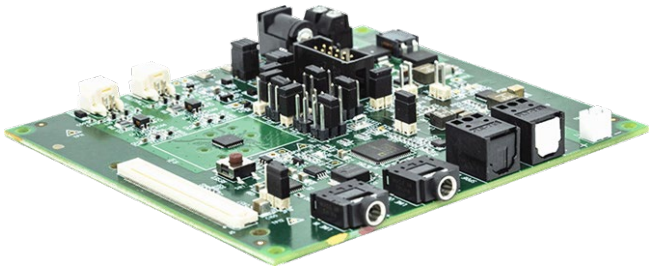


The Digital Signal taken from Audio Precision shown in Figure 4 is a digital 1 kHz or 11 kHz sine wave provided by the APx555 Audio Analyzer output to the AD2428 Transceiver WD1BZ Main Node board. Measurements are taken for both -1 dBFS and -40 dBFS input signals.

The Analog Signal connected back to Audio Precision shown in Figure 4 is connected to the input of the APx555 Audio Analyzer, and audio performance is measured. The A^{2}B bus is extended by inserting additional subordinate nodes (depicted by the *WD1BZ AD2428 Sub-Node* blocks in Figure 4), enabling measurements for the full set of sixteen subordinate nodes.

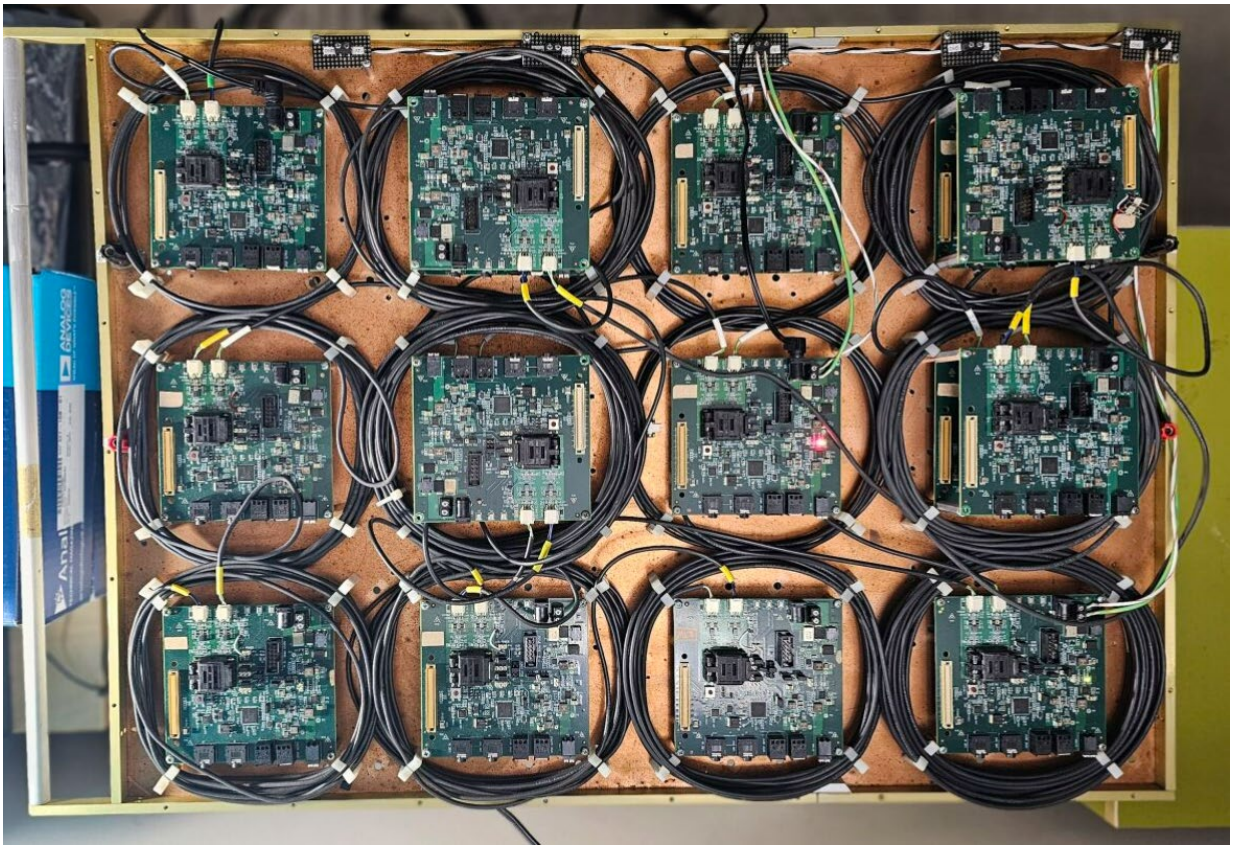
The test setup uses a combination of seventeen EVAL-AD2428WD1BZ evaluation kits (1 main node and 16 sub nodes). The evaluation kits were modified from standard customer evaluation kits using sockets for the AD2428 transceiver to allow for removing and installing different silicon revision parts (and in this case, slightly higher jitter performance parts). The evaluation kit is shown in Figure 5. Refer to the *EVAL-AD2428WD1BZ A2B Evaluation Board Manual*[3] for more information including setup and jumpers.

Figure 5 Hardware Configuration Uses Seventeen Total EVAL-AD2428WD1BZ Evaluation Kits



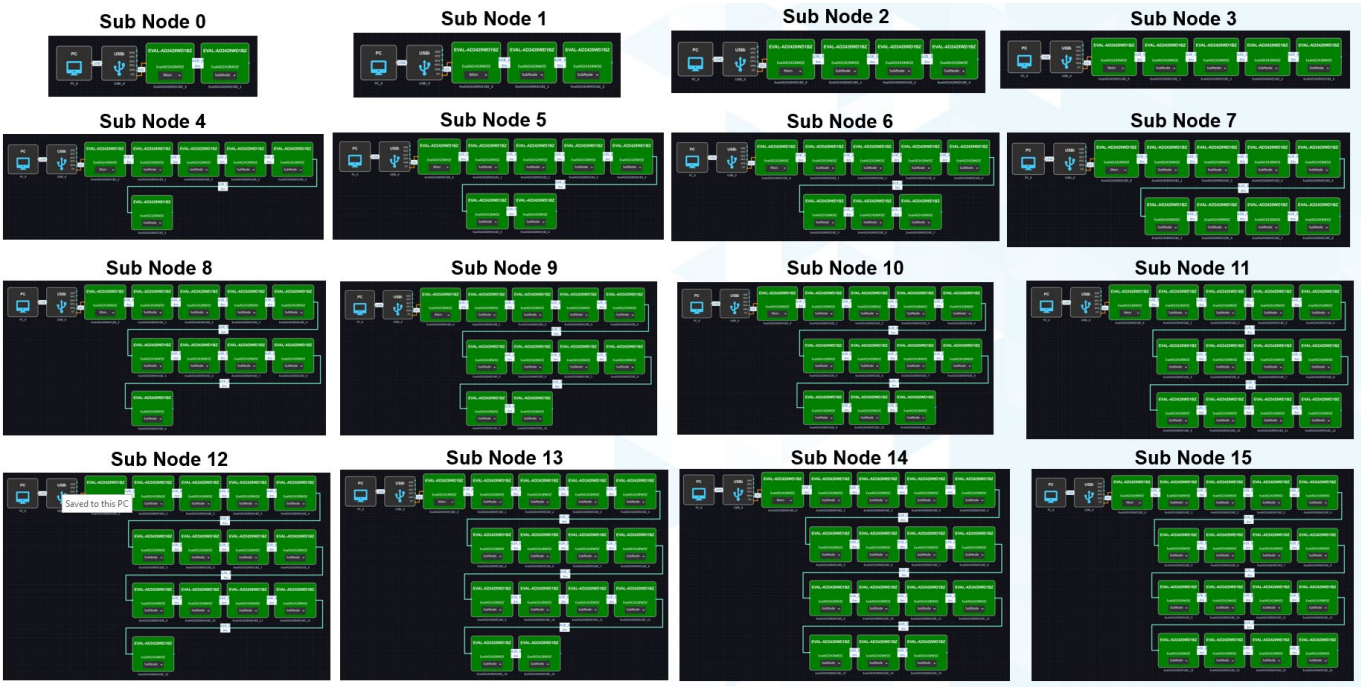
The A^2B network system is a combination local-powered WDZ subordinate nodes and bus powered WDZ subordinate notes shown. Depending on the configuration used to test the number of nodes, a local-powered sub node must be inserted in between every fourth or fifth node, so that sufficient voltage can drive the downstream boards. Figure 6 shows the test setup of 16 socketed EVAL-AD2428WD1BZ sub node boards.

Figure 6 THD+N Test setup using a network of EVAL-AD2428WD1BZ evaluation kits



Measurements taken during this test setup (Figure 6) included a single sub-node (sub[0], depicted as the single *WDZ Sub-Node* in Figure 4 with no *other WDZ Sub-Nodes* present) and then up to a maximum sixteen sub nodes (sub[0…15], where sub[n] is the *WDZ Sub-Node* in Figure 6. For each test setup, the measurement was taken at the last sub node, and all results are measured in THD+N dBrA, where A is set such that 0 dBrA = 0 dBFS. For each of the sixteen sub node tests, a customized SigmaStudio+ Schematic was developed for each sub node test, as shown below in Figure 7.

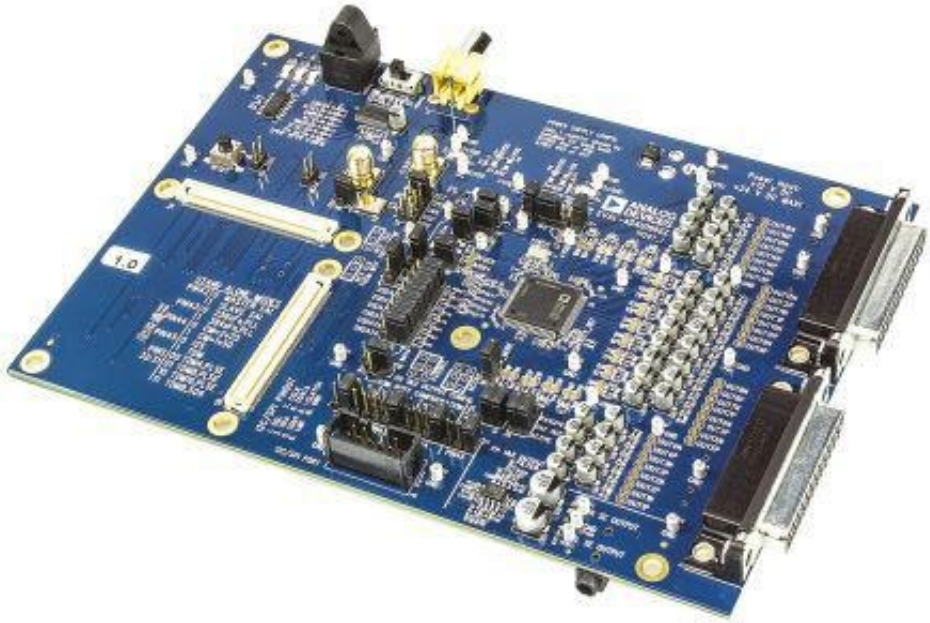
Figure 7 SigmaStudio+ 2.2 schematics for sub nodes 0 to 15 used for jitter analysis and THD+N testing



Selection of High-Quality DAC for THD+N Performance Testing

To examine the effects of accumulated jitter on an audio system, a high-quality DAC must be chosen to see when any negative audible artifacts are introduced because of the jitter introduced by consecutive A^2B sub nodes. The higher the dynamic range (SNR and THD+N), the lower the noise floor (baseline) in the FFT spectrum analysis. The Analog Devices DAC converter evaluation kit (EVAL-ADAU1966Z), shown in Figure 7 was chosen for this test. See *ADAU1966 16-Channel High Performance Differential Output, 192 kHz, 24-Bit DAC Datasheet*[8] for detailed set-up instructions.

Figure 8 EVAL-ADAU1966Z (left) DAC Evaluation Kits



For high THD+N performance specifications and testing, the ADAU1966 evaluation kit is one possible solution offered by Analog Devices to test the audio quality of an A^{2}B-connected system. This evaluation board was chosen specifically because its performance exceeds many other Analog Devices DACs currently sold in the market today. Figure 8 shows the excerpt related to the performance of the DAC taken from the ADAU1966 datasheet (THD+N at -98 dB, at 0 dBFS).

Figure 9 The ADAU1966 is one of the higher performing DACs from Analog Devices

ANALOG DEVICES

16-Channel High Performance Differential Output, 192 kHz, 24-Bit DAC

Data Sheet **ADAU1966**

FEATURES

118 dB DAC dynamic range and SNR
−98 dB THD + N

GENERAL DESCRIPTION

The ADAU1966 is a high performance, single-chip DAC that provides 16 digital-to-analog converters (DACs) with differen-

ANALOG PERFORMANCE SPECIFICATIONS

Specifications guaranteed at AVDDx = 5 V and an ambient temperature of 25°C. Supply voltages = AVDDx = 5 V, DVDD = 2.5 V, ambient temperature[1] (T_A) = 25°C, unless otherwise noted.

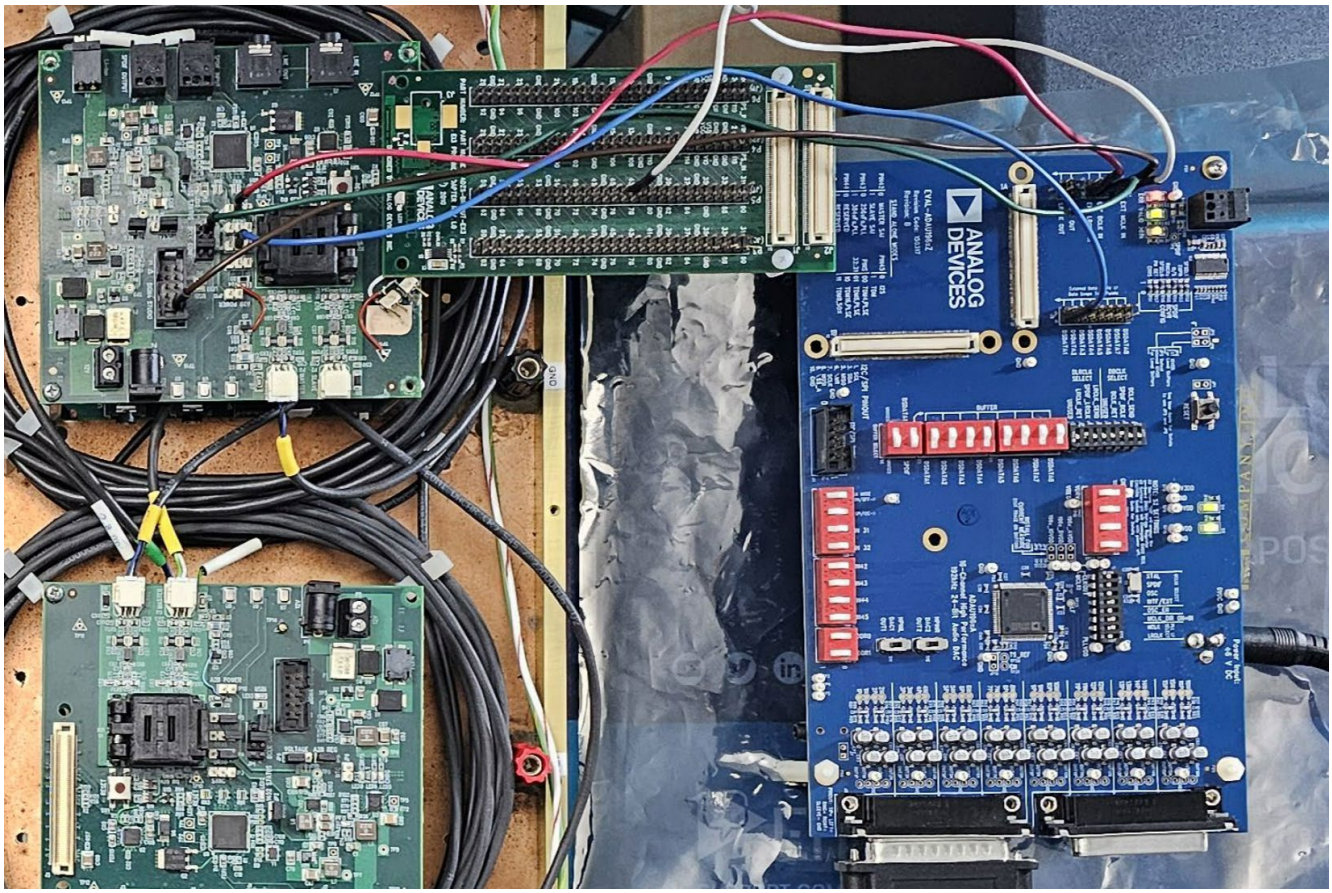
Table 1.

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
DIGITAL-TO-ANALOG CONVERTERS					
Dynamic Range	20 Hz to 20 kHz, −60 dB input				
No Filter (RMS)		105	115.5		dB
With A-Weighted Filter (RMS)		108	118		dB
Total Harmonic Distortion + Noise	0 dBFS		−90		dB
	Two channels running, −1 dBFS		−98		dB
	16 channels running, −1 dBFS		−98	−85	dB

Connecting the WDZB1 Sub Node Eval Kit I2S Output to the ADAU1966

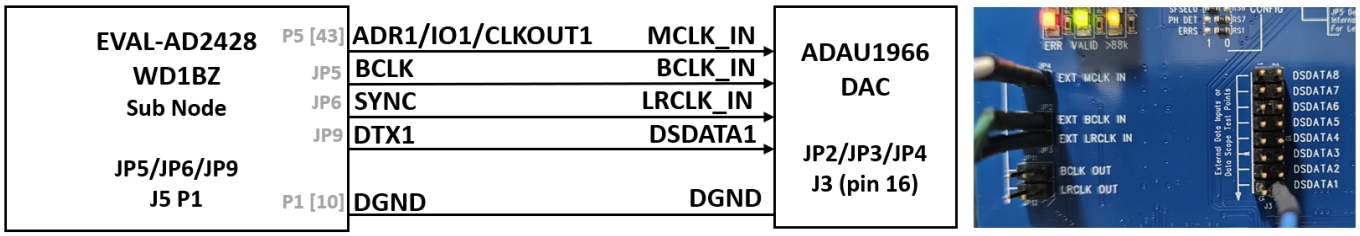
For our series of THD+N tests we expect to hear and measure the generated sine tone initially injected into the main AD2428 board streamed through the output of the last node. The last AD2428 sub node provides the audio output through its I^2S/TDM interface which we can further connect to the audio DAC. Many D/A converters typically require a master clock (MCLK) to clock the device, so we need to generate this clock signal as well from the recovered A2B clock. Specifically, when using the ADAU1966, the AD2428 has to provide a 12.288 MHz CLKOUT on the IO1/ADR1 pin, the 3.072 MHz BCLK, 48 kHz SYNC, and DTX serial audio data. The serial interface signals are provided to the JPx headers and the P5 expansion header and are shown in Figure 9. The IO1 pin is not easily accessible on the board through the any of the JPx jumpers. However, the IO1 pin is conveniently routed to the P5 expansion connector where we can access that clock pin. In order to tap into it, we can use the ADZS-BRKOUT-EX3 breakout board pin 43.

Figure 10 EVAL-AD2428WDZB1 P1 Header I^2S Signals



The block diagram for connecting the AD2428 WDZB1 evaluation kit I^2S pins to the ADAU1966 DAC evaluation board is shown in Figure 11, including the P5 and P10 headers and JP5/JP6/JP9 header pin connections described in the EVAL-AD2428WD1B1 evaluation kit schematics. Figure 11 also shows the connections with jumper wires for the ADAU1966 DAC board.

Figure 11 EVAL-AD2428WD1BZ Sub Node I²S Output Connections to the ADAU1966 DAC Eval Kits

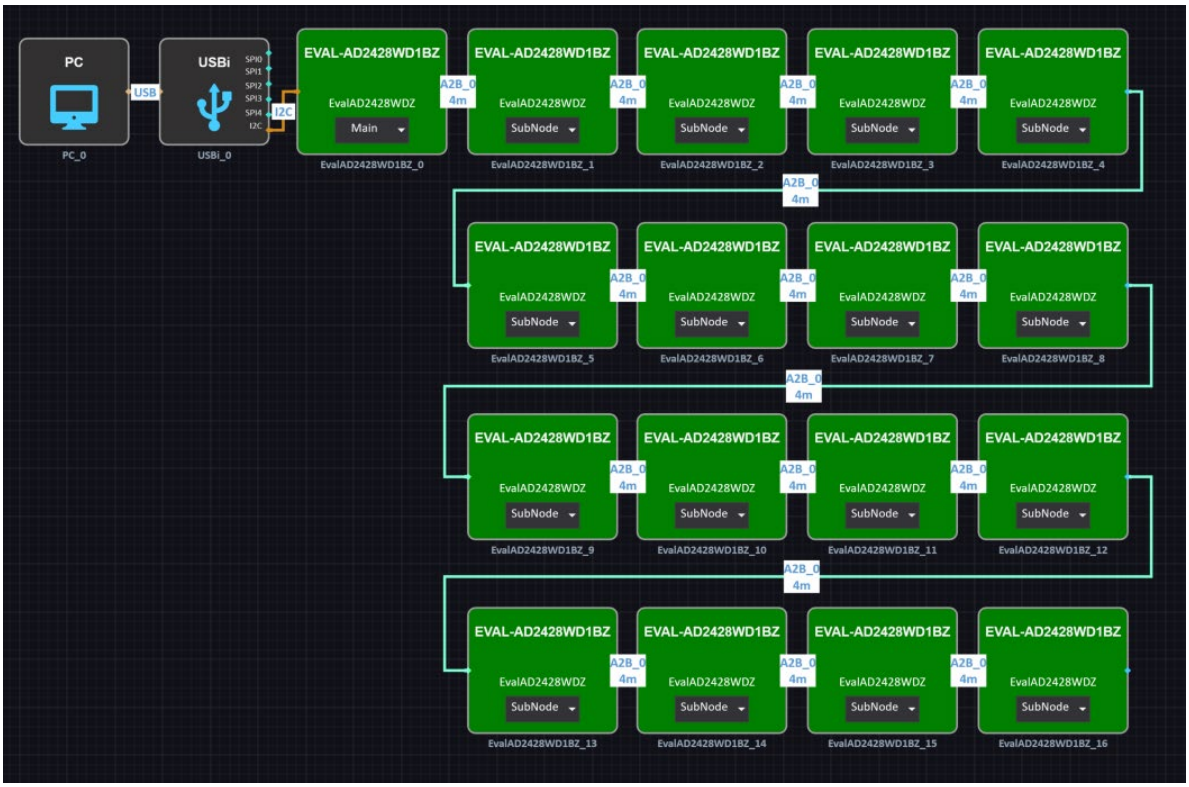


SigmaStudio+ Project Setup and Considerations

SigmaStudio+ version 2.2 and A²B plug-in version 1.3.1 was used to create the various AD2428 sub node test projects. The A2B plugin v.1.3.1 is the first supported release update which extended support for the AD2428 from 10 to 16 sub nodes. The SigmaStudio+ test project configuration shown in Figure 11 has one main AD2428 node and sixteen AD2428 sub node connections.

In the SigmaStudio+ schematic signal flow, the first node in the system shown is defined as the main node using an EVAL-AD2428WD1BZ board. This is followed by up to sixteen EVAL-AD2428WD1BZ sub node boards. In the test system shown in Figure 6, **sub nodes 0, 5 and 11 are locally-powered subs** using an external12-volt supply. The other sub nodes are bus-powered nodes in the system.

Figure 12 SigmaStudio+ 17-Node Configuration Used For the THD+N and Jitter Testing



Using SigmaStudio+ to generate a 12.288 MHz CLKOUT1 on ADR1/IO1 pin for the DAC's MCLK

To properly connect the WDZB1 board's I^2S interface to the DAC board, enable a clock source on the *last* sub node for the Master Clock (MCLK) of the DACs. By default, this clock is not enabled in SigmaStudio+. The P1 header provides the ADR1/IO1 signal, which is enabled by programming the CLKCFG register to turn on the clock and divide the 24.575 MHz A^2B clock by two to generate the needed 12.288 MHz MCLK for the DACs. Figure 13 and Figure 14 show the required register settings in SigmaStudio+.

Figure 13 SigmaStudio+ AD2428 Subnode Register Settings for CLKOUT1 and CLK1CFG Register

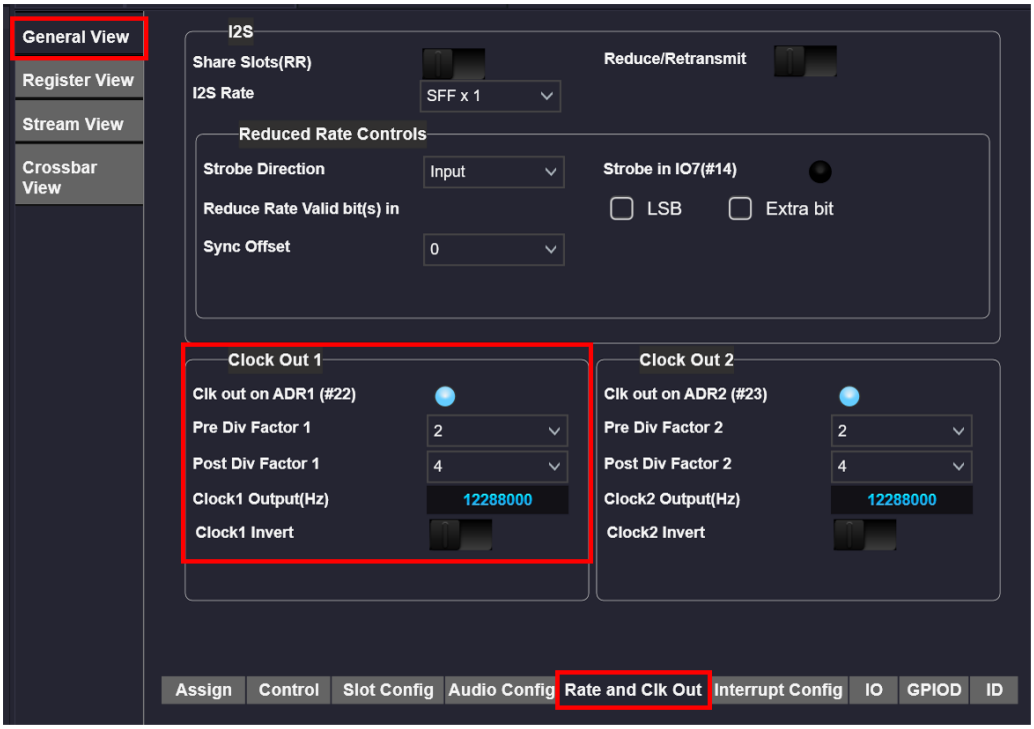


Figure 14 SigmaStudio+ CLK1CFG Register bits 0 and 7 are set to 1

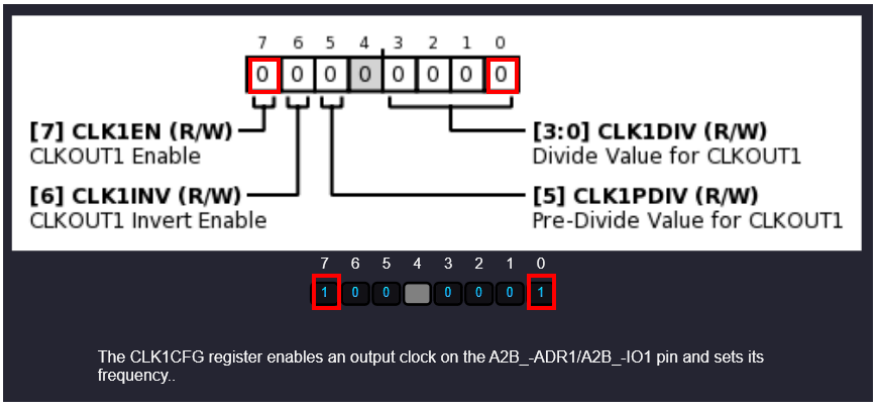
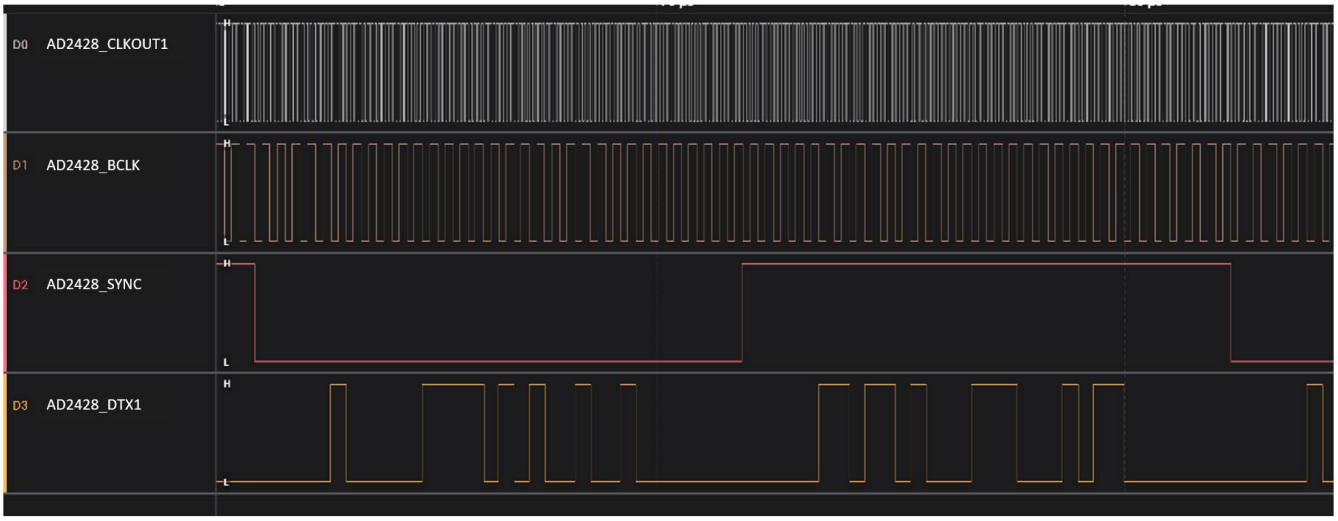


Figure 15 shows an analyzer capture of the 12.288 MHz MCLK supplied for the DAC (A2B's CLKOUT1) which becomes activated after programming the CLK1CFG register during discovery/initialization of the A2B system.

Figure 15 Capture of WD1BZ I2S output – 12.288 MHz MCLK, 3.072 MHz BCLK, and 48 kHz SYNC



Baseline Digital I2S THD+N Test

An audio analyzer such as the Audio Precision APx555 can be used for THD+N measurements for an A^2B system. The APx555's DSIO interface module is used to provide an I^2S source and sink into the system for the measurement. Two tests were performed, one testing the digital THD+N capability on the I2S inputs and outputs of the analyzer itself in a "loopback" mode, followed by a digital THD+N test for the entire I2S (i.e. I^2S input → A2B system → I^2S output) path through 16 x AD2428 sub nodes. Testing and results are described below.

APx555 DSIO Loopback of 1 kHz and 11 kHz Test Tones

As a first test, pure sine tones were injected and measured from the APx555 DSIO module's I2S transmitter back to the DSIO module's I2S receiver input. The FFT spectrum of the audio signals (Figure 16 and Figure 17) shows no harmonics or noise generated, except for quantization noise which were the smallest LSBs of the 32-bit audio data approximating -175 to -185 dB. The noise floor resulting from the quantization of the audio samples significantly less than the noise floor of the DAC and thus we are certain that this will not adversely affect the THD+N results for our DAC measurements.

Figure 16 1 kHz at -1 dBFS APx555 I²S Loopback

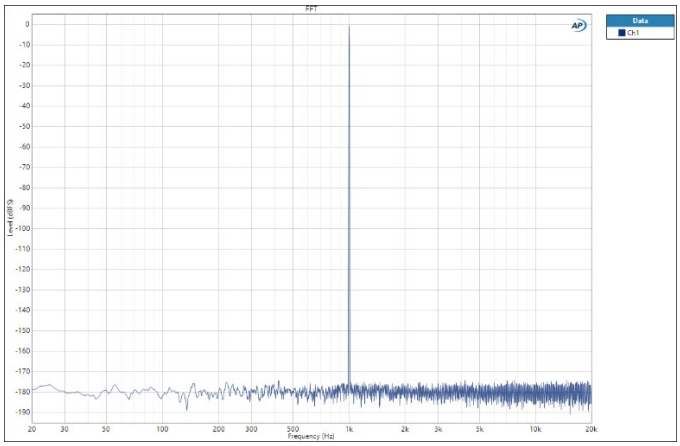
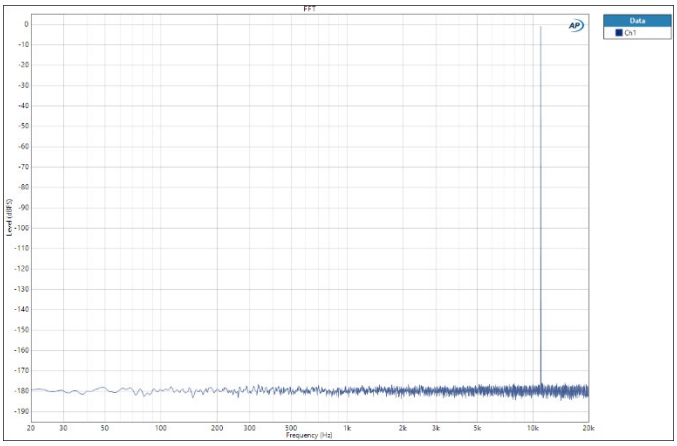


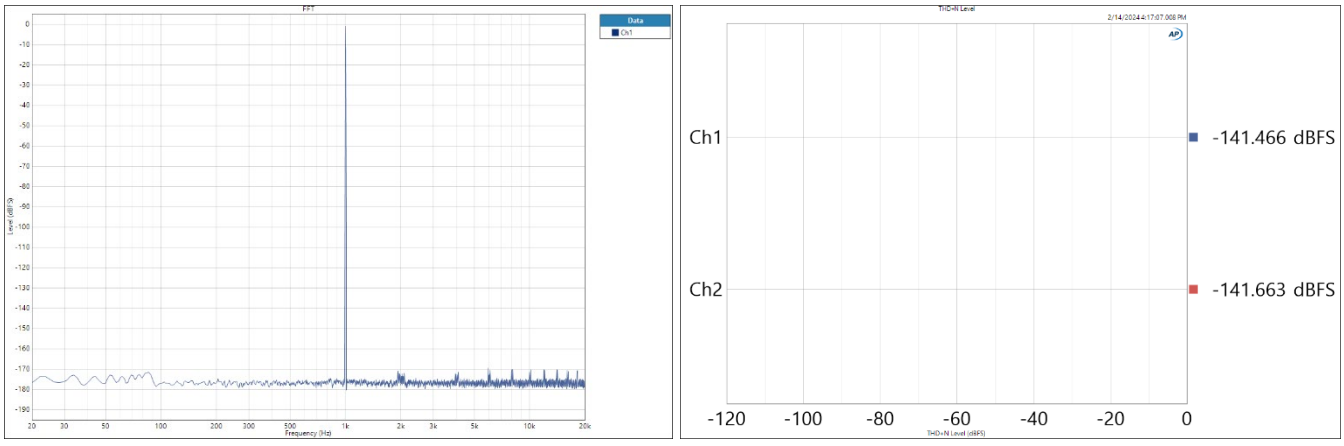
Figure 17 11 kHz at -1 dBFS APx555 I²S Loopback



1 kHz and 11 kHz Test Tones for THD+N Tests through a maximum 17-node configuration

Other measurements included the test tones injected from the APx555 DSIO Module's I²S transmitter on to the WDZB1 main board, and the DSIO Module I²S Receiver capturing the last sub node's I²S 24-bit digital data on the I²S output of the same 1 kHz tone streamed from the main node. The FFT and THD+N result is shown (Figure 18) for 1 kHz at -1 dBFS, measuring at approximately -142 dB. The THD+N due to quantization or bit errors on the digital signal is shown to be much less than typical results for the DAC converters by -20 to -30 dB. Thus, we can see here the digital path of the I2S data through the A2B system should not itself adversely affect the THD+N results for the DACs because effects on the audio quality due to quantization effects are significantly less than the noise floor for any typical automotive-quality audio converter such as the ADAU1966.

Figure 18 FFT and THD+N results for a 1 kHz sine tone at -1dBFS - I²S-to-I²S 16 sub node Loopback



DAC THD+N Performance Results with AD2428 Node Cumulative Jitter

Table 2 shows the results of the THD+N testing using the Audio Precision APx555 Audio Analyzer. The THD+N test parameters were set to A-weighted, band-limited frequency range from 20 Hz through 20 kHz. Table 2 displays the results for the ADAU1966 DAC. Test tones were measured at 1 kHz and 11 kHz at levels of -1 dBFS and -40 dBFS and THD+N results are shown along with the corresponding jitter measurement at each node. Each row corresponds to either the first baseline direct I^2S to DAC measurement or THD+N numbers for each A^2B sub-node. The sub node number corresponds to where the measurement was taken. For each test this was the last bus-powered sub-node on the A^2B bus.

Table 2 Results for AD2428WDZB1 Sub Node I^2S Output Connections to the ADAU1966

Sub node # I2S Output	APx555 Measured Jitter	ADAU1966 THD+N 1kHz -1 dBFS	ADAU1966 THD+N 1kHz -40 dBFS	ADAU1966 THD+N 11 kHz -1 dBFS	ADAU1966 THD+N 11 kHz -40 dBFS
I2S → DAC	75 ps	-97.83 dBrA	-113.12 dBrA	-112.54 dBrA	-115.84 dBrA
Sub 0	1.82 ns	-96.94 dBrA	-113.05 dBrA	-87.89 dBrA	-115.66 dBrA
Sub 1	2.14 ns	-96.32 dBrA	-113.02 dBrA	-86.44 dBrA	-115.56 dBrA
Sub 2	2.35 ns	-96.16 dBrA	-113.09 dBrA	-85.76 dBrA	-115.48 dBrA
Sub 3	2.64 ns	-95.82 dBrA	-112.94 dBrA	-85.49 dBrA	-115.46 dBrA
Sub 4	2.79 ns	-95.63 dBrA	-112.85 dBrA	-85.23 dBrA	-115.47 dBrA
Sub 5	2.94 ns	-95.42 dBrA	-113.10 dBrA	-85.04 dBrA	-115.47 dBrA
Sub 6	3.05 ns	-95.23 dBrA	-112.95 dBrA	-84.97 dBrA	-115.43 dBrA
Sub 7	3.13 ns	-95.07 dBrA	-113.11 dBrA	-84.88 dBrA	-115.35 dBrA
Sub 8	3.25 ns	-94.84 dBrA	-113.07 dBrA	-84.84 dBrA	-115.34 dBrA
Sub 9	3.41 ns	-94.47 dBrA	-113.02 dBrA	-84.82 dBrA	-115.27 dBrA
Sub 10	3.58 ns	-94.26 dBrA	-112.97 dBrA	-84.77 dBrA	-115.24 dBrA
Sub 11	3.70 ns	-94.16 dBrA	-113.07 dBrA	-84.75 dBrA	-115.22 dBrA
Sub 12	3.82 ns	-93.89 dBrA	-113.03 dBrA	-84.70 dBrA	-115.13 dBrA
Sub 13	3.94 ns	-93.63 dBrA	-112.77 dBrA	-84.61 dBrA	-115.02 dBrA
Sub 14	4.05 ns	-93.38 dBrA	-112.97 dBrA	-84.43 dBrA	-114.93 dBrA
Sub 15	4.19 ns	-92.71 dBrA	-113.13 dBrA	-84.18 dBrA	-114.79 dBrA

Analyzing the THD+N Results for the ADAU1966

The AD2428 parts used in the test setup at room temperature (25 °C) were taken from a sample lot which were screened and found to be higher than the average jitter numbers shown in Table 10 in the AD2428 datasheet [*AD2428 A2B Transceiver Datasheet*[1].

Examining the THD+N measurement results for the DAC, a significant degradation in audio performance was recorded (measured in dBrA units) with increasing accumulated jitter at higher nodes. These measured results using typical lot parts installed on Analog Devices A^2B evaluation boards showed approximately a 5 dB reduction in THD+N for the ADAU1966 DAC, at the maximum measurement RMS jitter of the A^2B bus. These results must be considered when reviewing the full-system measurements.

Based on measured results, it's apparent the ADAU1966 suffered a degradation in performance, however its THD+N calculation with higher accumulated jitter still performed much better to maintain compressed "CD-quality" compressed audio with greater than 90 dB. The ADAU1966 was found to provide better tolerance to jitter as compared to higher quality legacy Analog Devices converters such as the AD1955, perhaps due to a slightly-improved noise-immune design for the on-chip PLL. [Refer to *EE-452 – AD2437 A2B Transceiver Performance Testing* which includes testing for the AD1955 DAC]

For the audio engineer, the amount of acceptable degradation must be weighed, depending on the type of audio product to be designed. For a consumer or automotive audio design, anything between -90 to -95 dB THD+N might be acceptable or "*good enough*." However, for a professional audio product using converters with over 120 dB in dynamic range, this degradation can be unacceptable. Potential mitigation strategies (discussed later in this application note) may need to be employed in the A^{2}B system to prevent adverse effects on the performance of the audio converters, which can derive their I^{2}S or TDM serial clocks and frame syncs from the A^{2}B bus recovered clock.

Additionally, the *AD2426/AD2427/AD2428 A2B Transceiver Datasheet*[1]. does specify worst-case jitter values. It is possible to see jitter values up to 60 to 70% higher than the typical TIE$_{RMS}$ value. Expect an additional reduction in THD+N than the test results showed.

FFT Spectrum Diagrams for -1/-40 dBFS using 1 kHz/11 kHz Test Tones

Figure 21 through Figure 68 show different various FFT Spectrum plots from four THD+N tests. It measures the performance of the audio quality while connected directly to the ADI D/A Converter Evaluation Kits to establish a best case expected performance value (the *baseline* measurement). After the baseline plots, measured frequency spectrum plots are shown for I^{2}S audio output stages for sub node zero, sub node four, sub node eight, sub node twelve, and sub node sixteen for both DACs.

ADAU1966 1 kHz -1 dBFS Test Results

Figure 19 shows the FFT spectrum analysis plots for the baseline direct I^{2}S connection to the ADAU1966 DAC from the DSIO output of the Audio Precision APx555, followed by spectrum plots (Figure 20 through Figure 24) for the A^{2}B bus configurations of sub node 0, 4, 8, 12, and 15 where the A^{2}B sub node I^{2}S output is fed to the ADAU1966 DAC. The results are based on a 1 kHz input sine wave at-1 dBFS.

Figure 19 1 kHz, -1 dBFS APx555 to ADAU1966

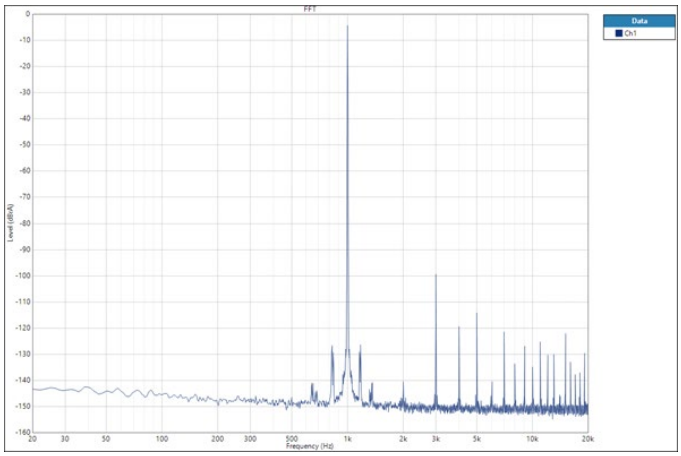


Figure 20 1 kHz, -1 dBFS S0 I^{2}S to ADAU1966

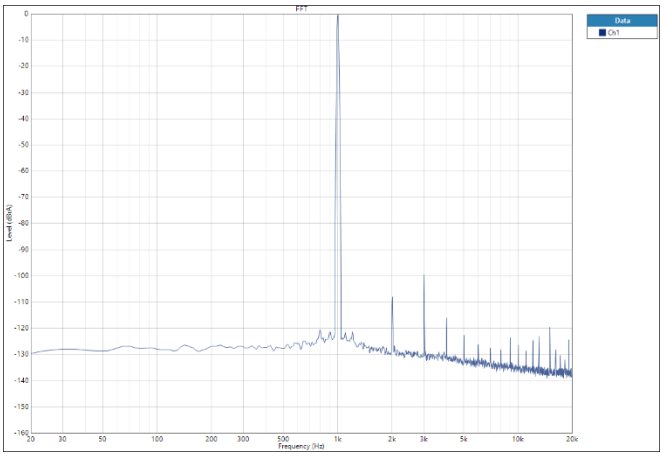


Figure 21 1 kHz, -1 dBFS S4 I²S to ADAU1966

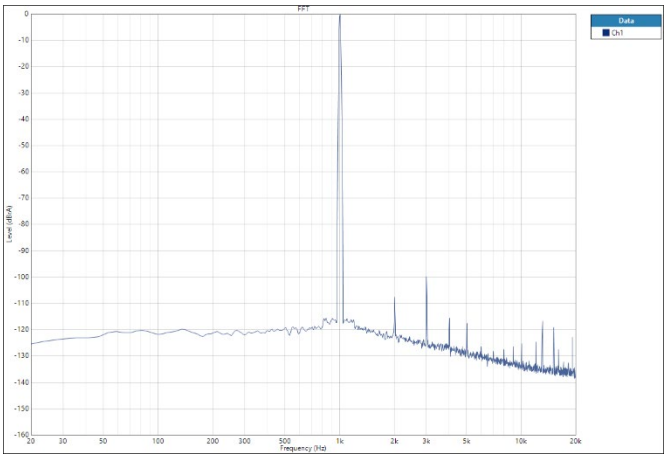


Figure 22 1 kHz, -1 dBFS S8 I²S to ADAU1966

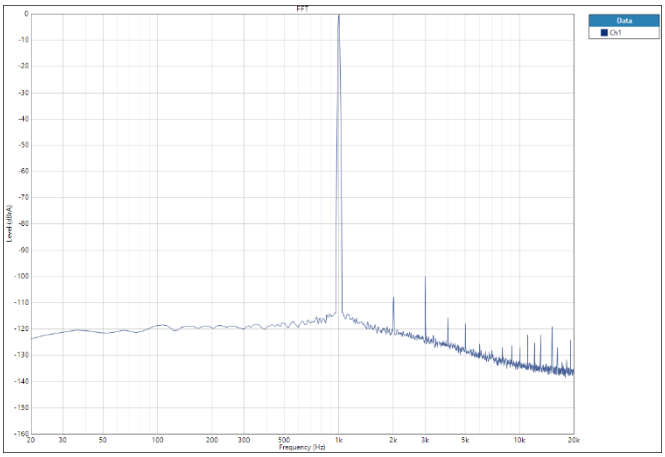


Figure 23 1 kHz, -1 dBFS S12 I²S to ADAU1966

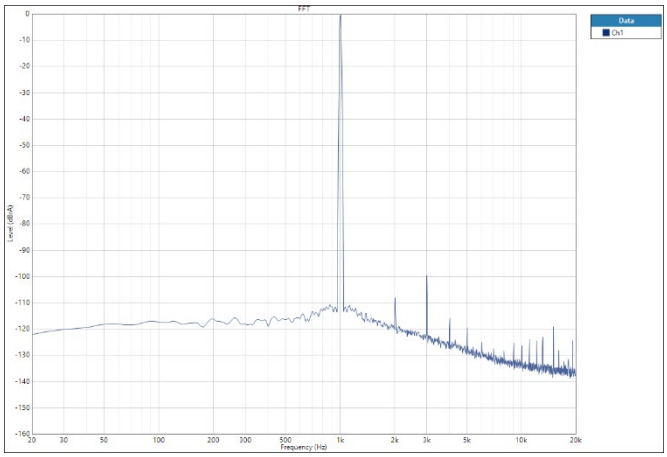


Figure 24: 1 kHz, -1 dBFS S15 I²S to ADAU1966

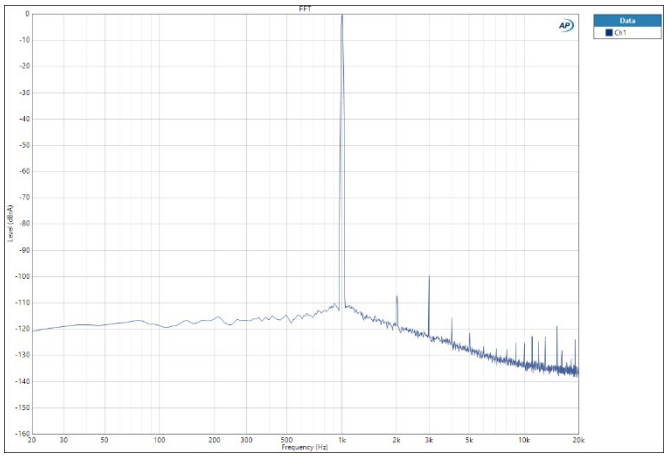


Figure 19 through Figure 24 Observations. The noise floor (measure of the noise density in dBm/Hz or the noise power in a signal of 1 Hz bandwidth) is raised significantly in the first sub node, and it is raised incrementally further as the number of nodes increases. Some higher frequency harmonics and noise are generated at higher-numbered nodes which do not exist for the initial baseline results, especially visible in the last node (sub node 15) for the ADAU1966.

ADAU1966 1 kHz -40 dBFS Test Results

Figure 25 shows the FFT spectrum analysis plots for the baseline direct I²S connection to the ADAU1966 DAC from the DSIO output of the Audio Precision APx555, followed by A²B bus configurations of sub node 0, 4, 8, 12, and 15 where the A²B sub node I²S output is fed to the ADAU1966 DAC (Figure 35 through Figure 44). The results are based on a 1 kHz input sine wave at -40 dBFS.

Figure 25 1 kHz, -40 dBFS APx555 to ADAU1966

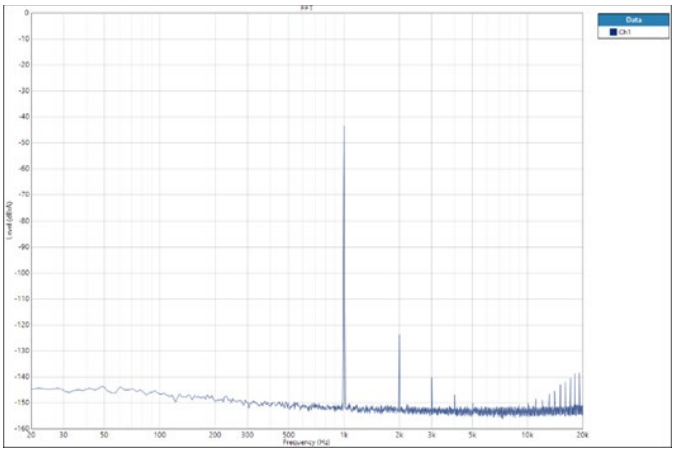


Figure 26 1 kHz, -40 dBFS S0 I²S to ADAU1966

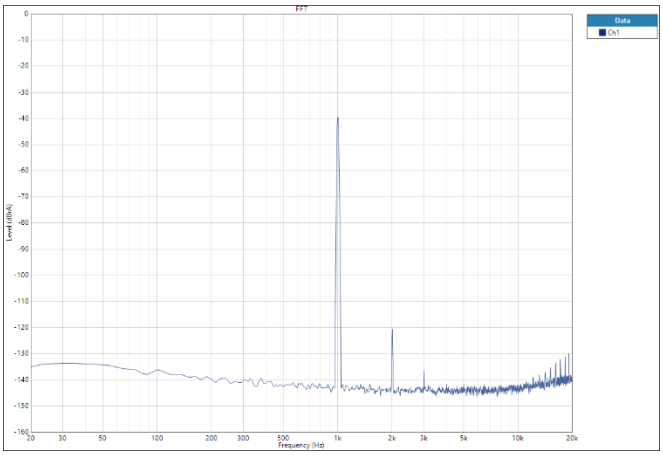


Figure 27 1 kHz, -40 dBFS S4 I²S to ADAU1966

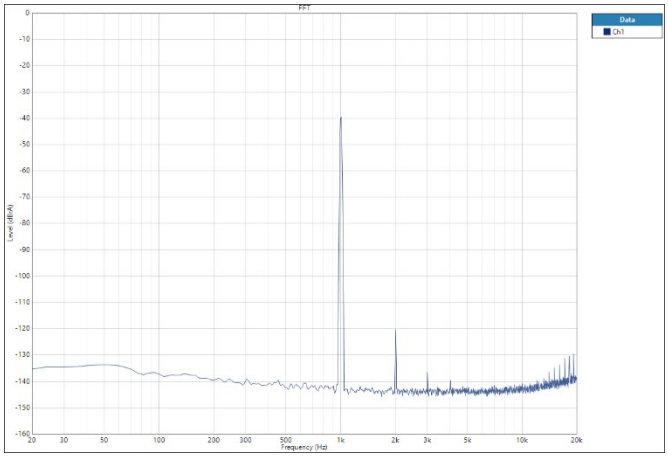


Figure 28 1 kHz, -40 dBFS S8 I²S to ADAU1966

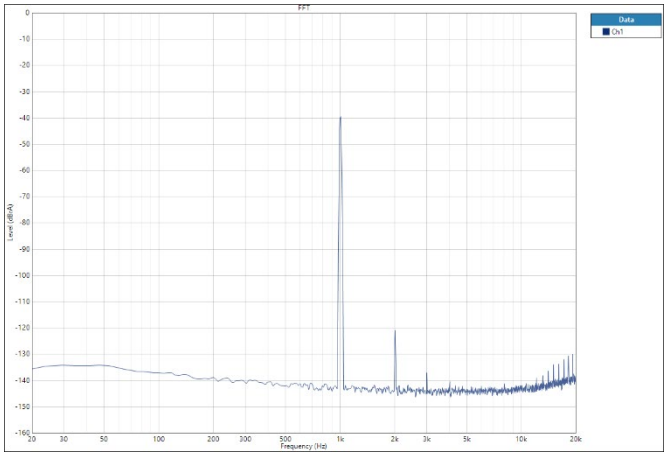


Figure 29 1 kHz, -40 dBFS S12 I²S to ADAU1966

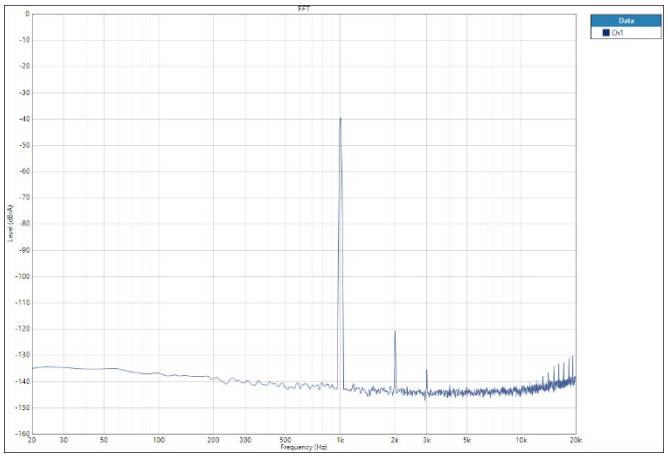


Figure 30 1 kHz, -40 dBFS S15 I²S to ADAU1966

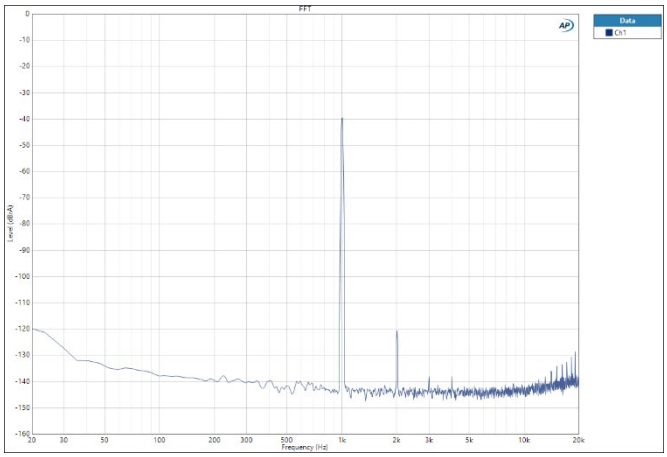


Figure 25 Through Figure 30 Observations. The noise floor at -40 dB is not much different than the baseline case. Perhaps worst case is two to three dB increase in noise floor for higher-numbered nodes. An observation is that the baseline spectrum for the ADAU1966 shows some high frequencies noise or harmonics between 10 kHz and 20 kHz, and the noise vanishes when the I^2S data is received from the A^2B bus and higher-jitter clock.

ADAU1966 11 kHz Test Results at -1 dBFS

Figure 31 shows the FFT spectrum analysis plots for the baseline direct I^2S connection to the ADAU1966 DAC from the DSIO output of the Audio Precision APx555, followed by A^2B bus configurations (Figure 32 through Figure 36) for sub node zero, sub node four, sub node eight, sub node twelve, and sub node fifteen where the A^2B sub node I^2S output is fed to the ADAU1966 DAC. The results are based on a 11 kHz input sine wave at -1 dBFS.

Figure 31 11 kHz, -1 dBFS APx555 to ADAU1966

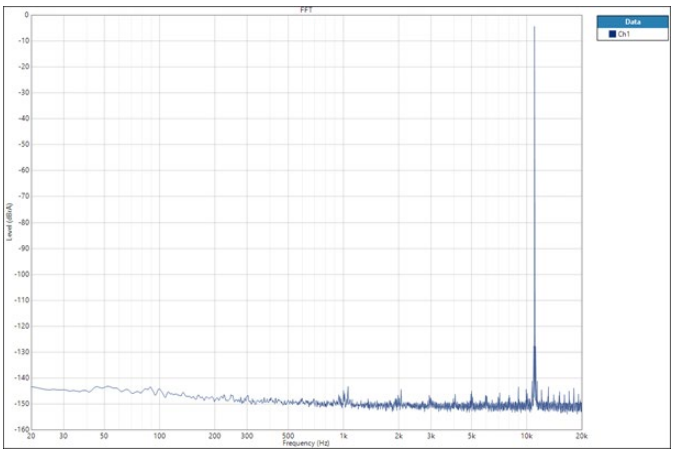


Figure 32 11 kHz, -1 dBFS S0 I^2S to ADAU1966

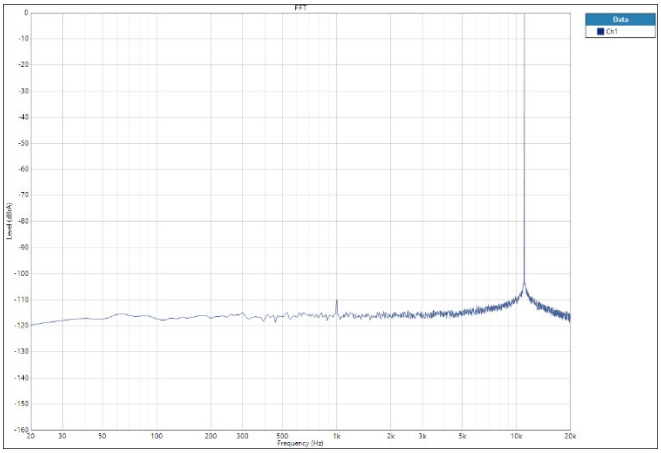


Figure 33 11 kHz, -1 dBFS S4 I^2S to ADAU1966

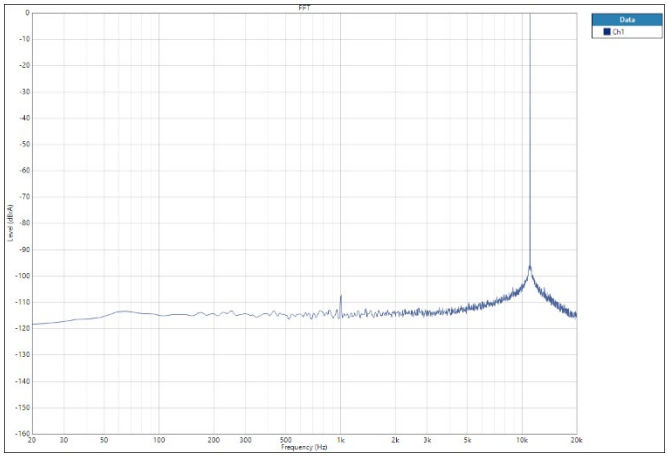


Figure 34 11 kHz, -1 dBFS S8 I^2S to ADAU1966

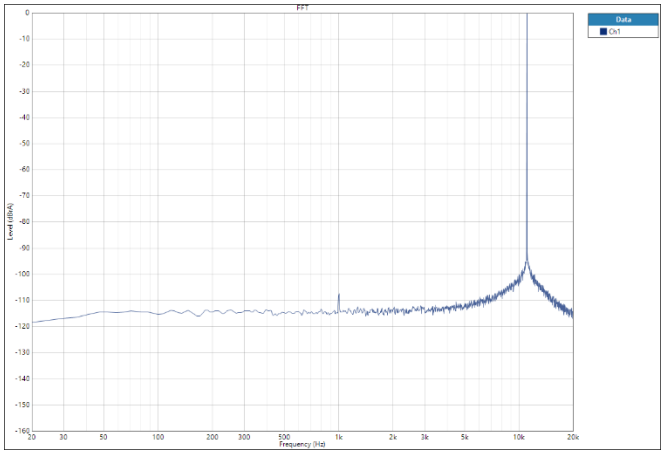


Figure 35 11 kHz, -1 dBFS S12 I²S to ADAU1966

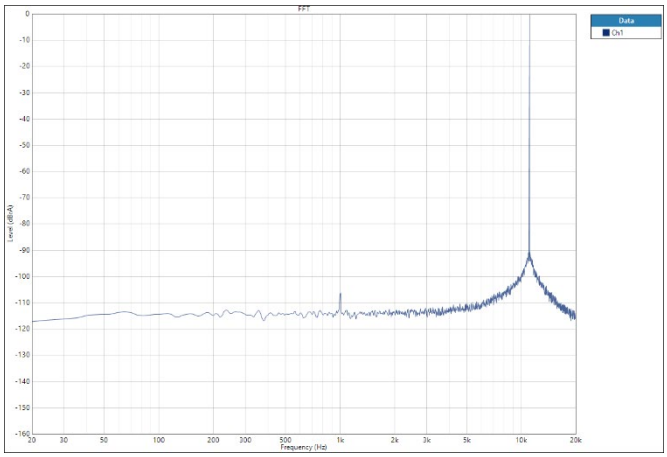


Figure 36 11 kHz, -1 dBFS S15 I²S to ADAU1966

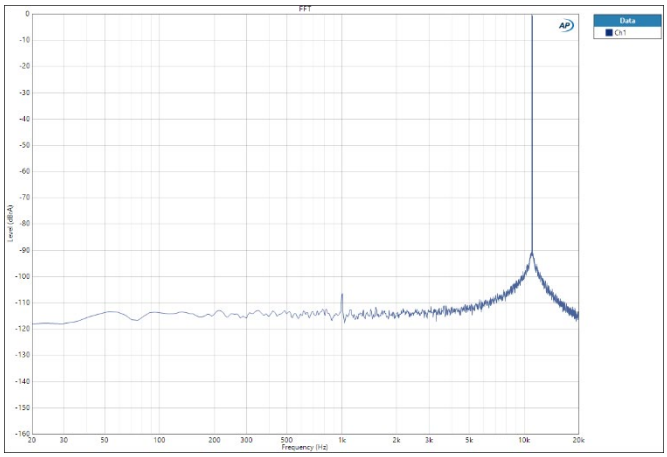


Figure 31 Through Figure 36 Observations. Once again, at -1 dB the noise floor is raised significantly, especially around the 11 kHz signal, and is raised further for the higher-numbered nodes. The low frequency content again eventually vanishes into the noise floor. The ADAU1966 DAC shows harmonic or tone at 1 kHz. The accumulated jitter causes this mysterious tone to display approximating -105 dB to -115 dB.

ADAU1966 11 kHz Test Results at -40 dBFS

Figure 37 shows the FFT spectrum analysis plots the baseline direct I²S connection to the ADAU1966 DAC from the DSIO output of the Audio Precision APx555, followed by A²B bus configurations (Figure 38 through Figure 42) of sub node zero, sub node four, sub node eight, sub node twelve, and sub node fifteen where the A²B sub node I²S output is fed to the DAC. The results are based on a 11 kHz input sine wave at -1 dBFS.

Figure 37 11 kHz, -40 dBFS AP I²S out to ADAU1966

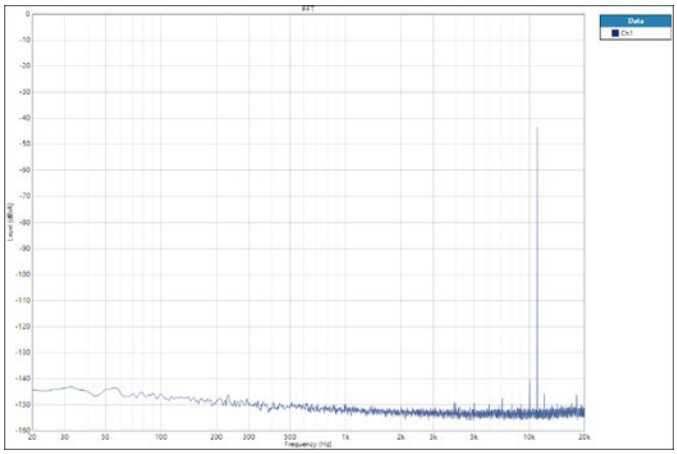


Figure 38 11 kHz, -40 dBFS S0 I²S to ADAU1966

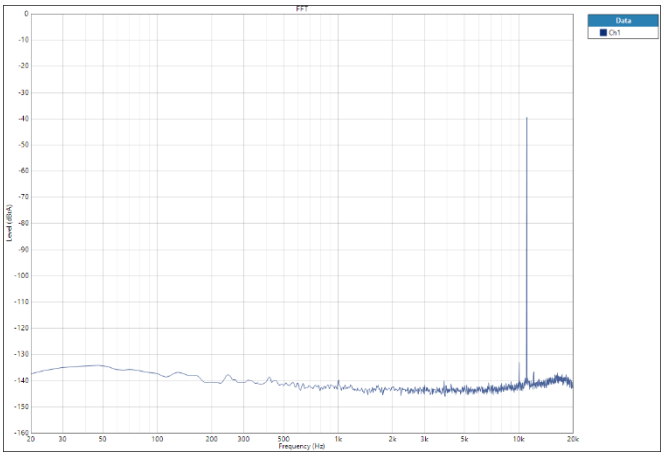


Figure 39 11 kHz, -40 dBFS S4 I²S to ADAU1966

Figure 40 11 kHz, -40 dBFS S8 I²S to ADAU1966

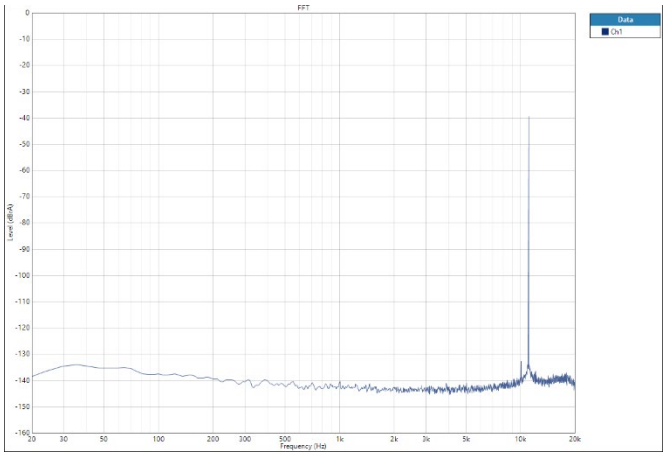


Figure 41 11 kHz, -40 dBFS S12 I²S to ADAU1966

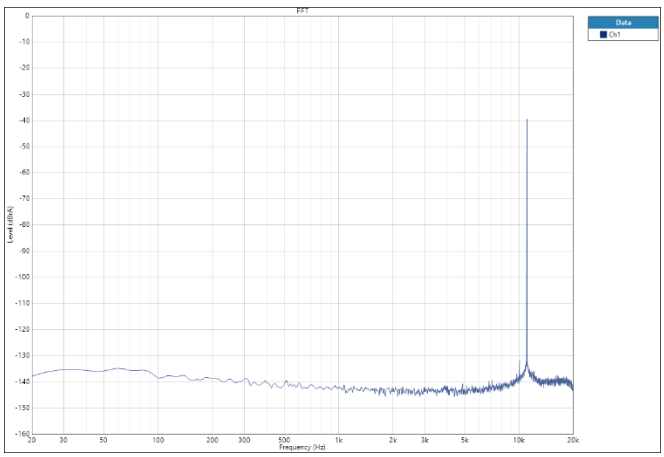
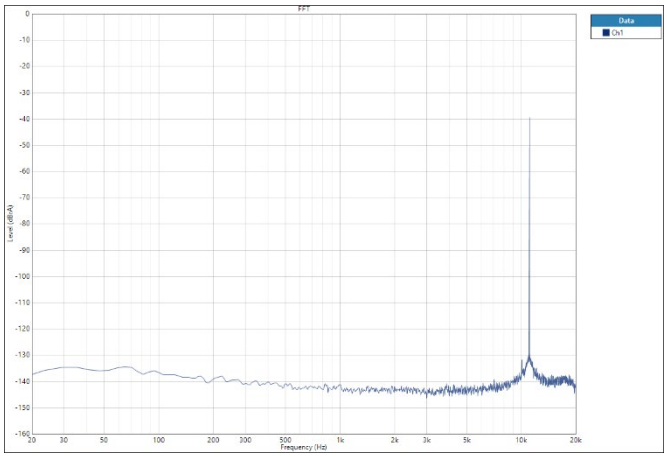


Figure 42 11 kHz, -40 dBFS S15 I²S to ADAU1966



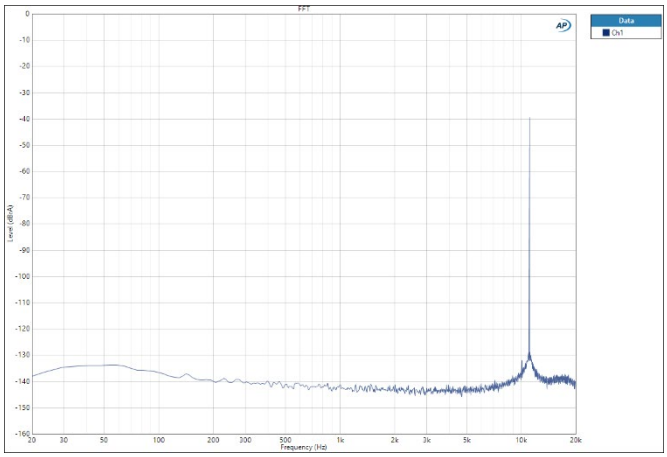


Figure 37 through Figure 42 Observations. The noise floor is not much different (slightly higher) than the baseline case with degradation of THN+N is approximately -1.5 dB for higher-numbered nodes. For the ADAU1966 some of the frequencies between 5K to 20K are raised by approximately 5 dB, while others vanish as compared to baseline I²S case.

Strategies for Mitigation of Audio Degradation

For system designs using ADCs and DACs, onboard PLLs with more robust specifications ensure a cleaner clock signal for the converters, reducing the performance degradation of the audio signals.

However, when the DAC performance is still affected by the accumulated jitter in the recovered I²S bit and word clocks, possible methods to mitigate the audio degradation due to accumulated jitter effects are:

- Perform a sample rate conversion at the DAC to align the samples to a clean clock signal. This is possible because the maximum accumulation of jitter is measured in single-digit nanoseconds, but the digital data remains correct. Resampling to a clean, low-jitter clock reference for the master I²S

output of the rate converter eliminates the THD+N degradation and reduces the clock jitter on the serial bus to a range of 100 to 200 picoseconds.

- Use a low-bandwidth PLL or Jitter Attenuator (JA) device to remove the upper-frequency components of the jitter on the recovered clock. This provides a similar performance benefit as the resampling data to a different master clock source. Jitter can be reduced to less than 100 picoseconds.

Solution One-Use a HW ASRC to Improve DAC THD+N Performance on Sub Nodes

Asynchronous sample rate conversion (ASRC) is a technique normally used to convert between audio streams running at different sample rates, or to perform sample rate conversion between audio streams running at the same sample rate but with different clock domains, such as a 48-kHz to 48-kHz rate conversion. An ASRC will convert the sample rate of a digital signal without locking the output clock to the input clock. The output clock can clock the audio data free from jitter effects of the input clock for the audio data samples. ASRCs can modify the sampling frequence in small increments over time without producing audible artifacts. Thus, we can utilize this rate conversion to transfer the same audio data to a new clock domain, and specifically in the case of a A^2B-recovered clock, move the resampled data synchronized to a new I^2S master with significantly less jitter (free) master clock, bit clock, and frame sync.

This next test case uses the same A^2B 16-node system, but with a HW ASRC inserted between the last node and the DAC board. For this test, a separate *EVAL-ADAU1452 Obsolete Evaluation Board*[26] was used, primarily because the EVAL-AD2428WDZ board's ADAU1452 does not provide additional dedicated serial port output headers or jumpers to easily access separate SPORT output pins.

Figure 43 shows the functional block diagram defining the necessary connections required between the EVAL-AD2428WDZ sub node, the ADAU1452MINIZ board, and the ADAU1966 eval kit. The ADAU1452MINIZ board is configured to receive I^2S audio on SPORT2 and transmit rate-converted audio on SPORT3. These signals are accessible on header J3 and the various test points provided on the board. Figure 44 shows the actual test setup between the three boards using jumper wires to connect the I^2S pins between the A^2B transceiver, the DSP, and the DAC.

Figure 43 Test Setup with Sub Node 15 Connection to ADAU1452's ASRC0 to ADAU1966 DAC

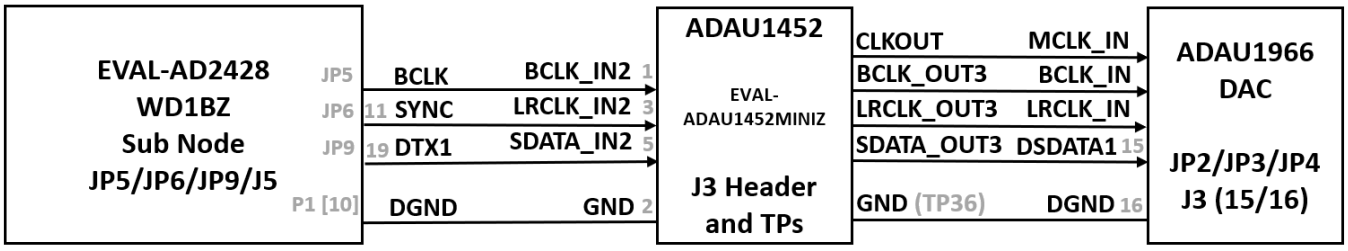
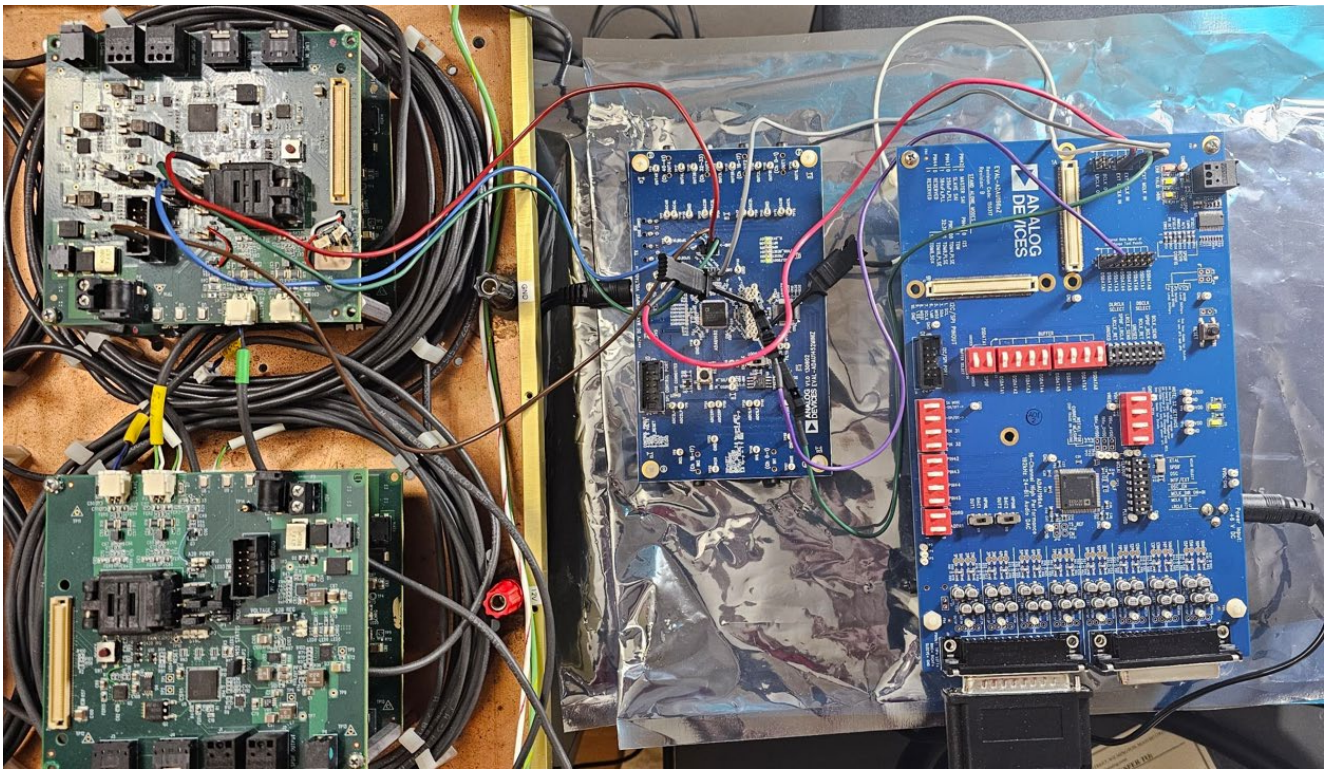
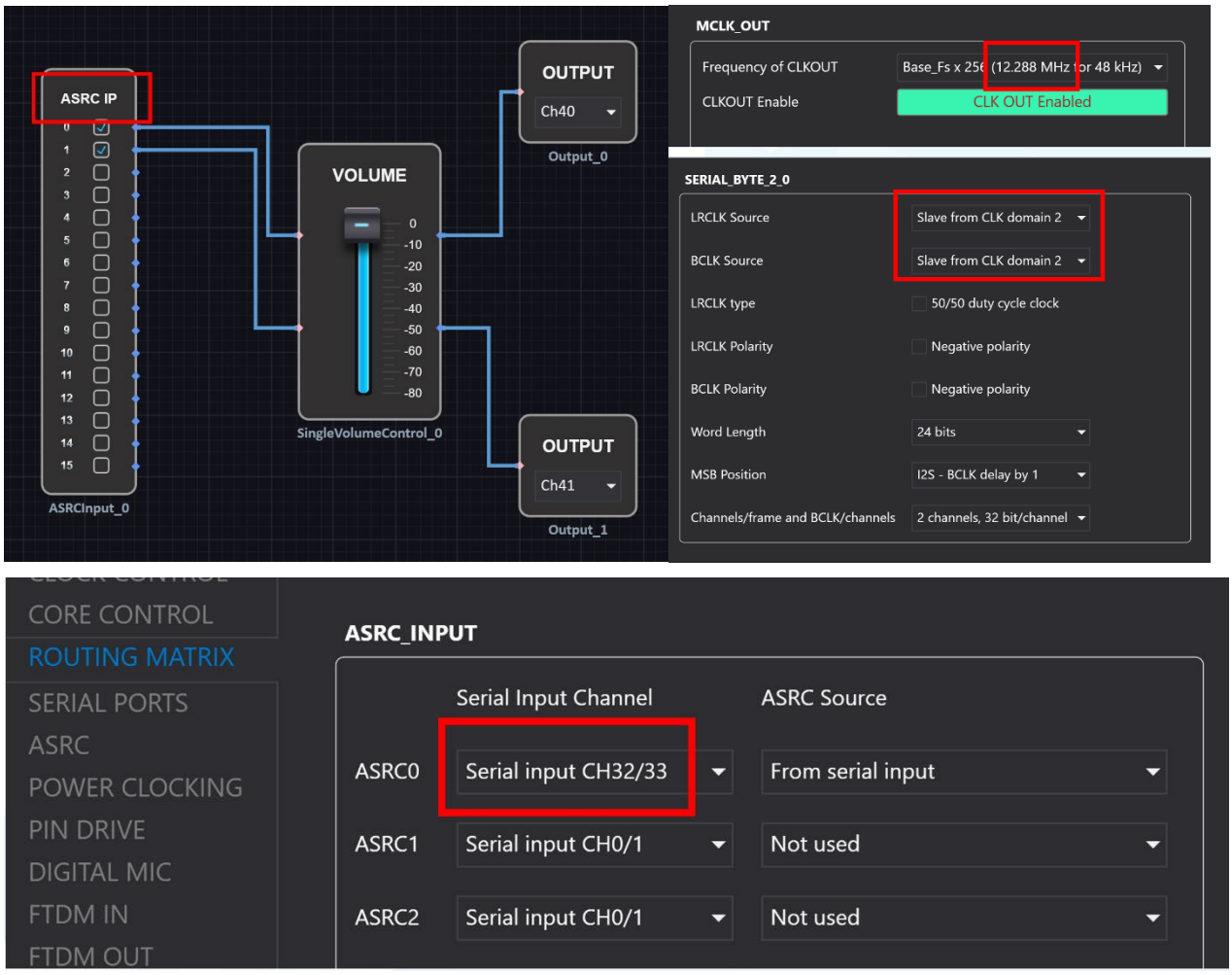


Figure 44 Test Setup with Sub Node 15 Connection to ADAU1452's ASRC0 to ADAU1966 DAC



To configure the ADAU1452 evaluation kit, a SigmaStudio+ 2.2 SigmaDSP schematic project was created for a simple passthrough of audio data coming from SPORT2, which is connected to the last sub node AD2428 WDZ board's I^2S output. The I^2S data on SPORT2 is connected to ASRC0. The output of ASRC0 is fed to the ADAU1452 Sigma300 DSP core, performing a simple audio *talkthru* out to the channels associated with SPORT3 (configured as an I^2S master). Lastly, we configure the CLKOUT to generate a 12.288 MHz. This is needed to drive the MCLK inputs for the ADAU1966. The project settings in Figure 45 configure the ASRC0, SPORT2, SPORT3, and CLKOUT/MCLK. The settings are the I^2S slave SPORT2 input to SPORT3 Master I^2S output using ASRC0, plus generating a 12.288 MHZ Master CLKOUT for the ADAU1966 MCLK Source.

Figure 45 SigmaStudio+ ADAU1452 Project Settings



DAC THD+N Performance Results with ADAU1452 Integrated HW ASRC

Table 3 below show the measured results when adding the ADAU1452 ASRC (asynchronous sample rate converter) block in the signal path between the last "nth" sub node in the A2B system and the connection to the ADAU1966 DAC. First in the test, the jitter was measured on the output of the ADAU1452 I2S output after passing through the HW ASRC block. Compared to the previous measured accumulated jitter results shown in Table 2, the measured jitter at the ADAU1452 DSP's ASRC/I2S out path showed a significantly reduction from the accumulated A2B bus jitter to now only about 170 to 230 picoseconds of jitter for BCLK and SYNC clocks. The THD+N measurements in the table below shows that the reduced jitter resulted in a much-improved THD+N performance matching the datasheet typical THD+N value for the ADAU1966. This was regardless of the addition of any higher number of nodes in the A2B system. The results are almost identical for each node. Thus, the use of the ASRC demonstrates the benefit of reduced I2S clock jitter which also provides optimal THD+N performance for the D/A converters – once sensitive to accumulated clock jitter affects. The results below show the DAC performance is equal to the datasheet typical THD+N numbers of -98 dB at for the ADAU1966 using a tone generated at 1 kHz and -1 dBFS.

Table 3. ADAU1966 THD+N Results with ASRC Inserted Between Sub Node [n] Path to DAC

Sub node # I2S Output	APx555 Measured Jitter	ADAU1966 THD+N 1kHz -1 dBFS	ADAU1966 THD+N 1 kHz -40 dBFS	ADAU1966 THD+N 11 kHz -1 dBFS	ADAU1966 THD+N 11 kHz -40 dBFS
I2S → DAC	70 ps	-98.34 dBrA	-113.12 dBrA	-112.54 dBrA	-115.84 dBrA
Sub 0	179 ps	-98.25 dBrA	-113.06 dBrA	-112.02 dBrA	-115.77 dBrA
Sub 1	217 ps	-98.37 dBrA	-112.96 dBrA	-111.91 dBrA	-115.96 dBrA
Sub 2	211 ps	-98.43 dBrA	-113.01 dBrA	-111.89 dBrA	-115.78 dBrA
Sub 3	212 ps	-98.34 dBrA	-112.94 dBrA	-111.82 dBrA	-115.87 dBrA
Sub 4	223 ps	-98.24 dBrA	-112.83 dBrA	-111.93 dBrA	-115.74 dBrA
Sub 5	214 ps	-98.29 dBrA	-113.05 dBrA	-111.75 dBrA	-115.96 dBrA
Sub 6	213 ps	-98.41 dBrA	-113.07 dBrA	-111.95 dBrA	-115.98 dBrA
Sub 7	218 ps	-98.32 dBrA	-113.03 dBrA	-111.73 dBrA	-115.89 dBrA
Sub 8	210 ps	-98.53 dBrA	-113.11 dBrA	-111.97 dBrA	-115.93 dBrA
Sub 9	219 ps	-98.42 dBrA	-113.02 dBrA	-112.04 dBrA	-115.81 dBrA
Sub 10	225 ps	-98.26 dBrA	-112.94 dBrA	-111.78 dBrA	-115.72 dBrA
Sub 11	230 ps	-98.35 dBrA	-113.09 dBrA	-111.91 dBrA	-115.85 dBrA
Sub 12	221 ps	-98.33 dBrA	-112.95 dBrA	-111.72 dBrA	-115.89 dBrA
Sub 13	225 ps	-98.39 dBrA	-113.06 dBrA	-111.81 dBrA	-115.81 dBrA
Sub 14	222 ps	-98.41 dBrA	-112.98 dBrA	-111.83 dBrA	-115.76 dBrA
Sub 15	226 ps	-98.28 dBrA	-113.08 dBrA	-111.81 dBrA	-115.67 dBrA

ADAU1966 -1 dBFS Test Results with ADAU1952 ASRC0 Added Before DAC

Figure 46 through Figure 51 are the supporting FFT spectrum analysis plots for the direct I^2S connection of the ADAU1966 DAC from the DSIO output of the Audio Precision APx555, plus the sub node zero, sub node four, sub node eight, sub node twelve, and sub node fifteen A^2B bus configurations. All figures depict the performance with a 1 kHz input sine wave at -1 dBFS.

Figure 46 1 kHz, -1 dBFS AP I²S Out to ADAU1966

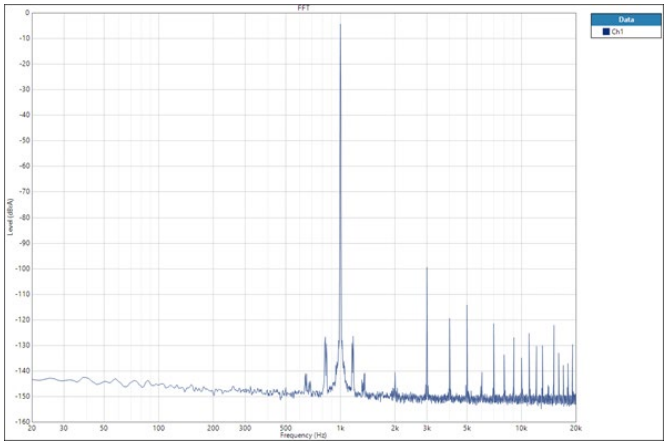


Figure 47: 1 kHz, -1 dBFS S0 to ASRC to ADAU1966

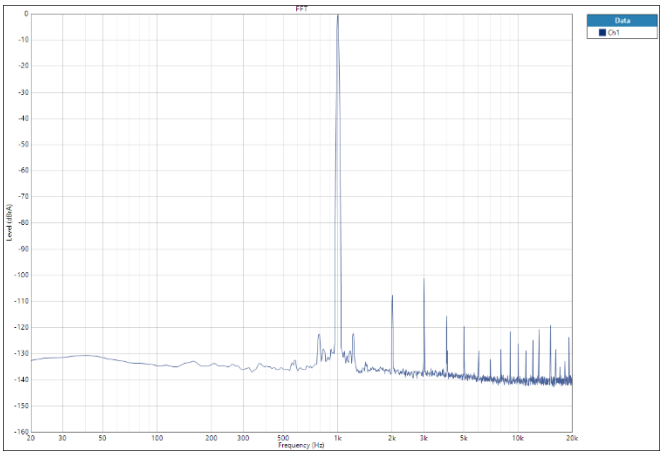


Figure 48: 1 kHz, -1 dBFS S4 to ASRC to ADAU1966

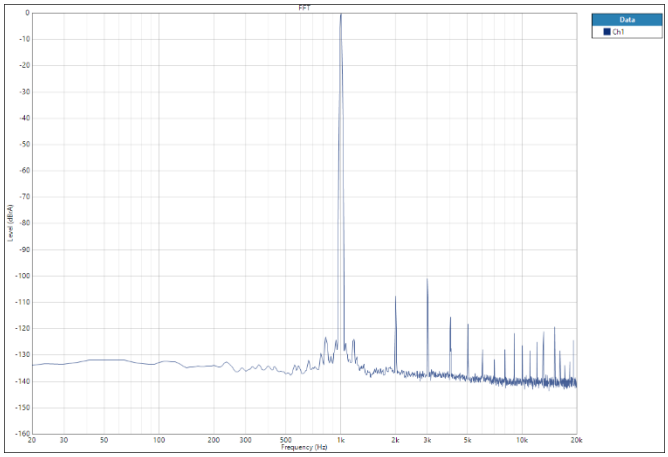


Figure 49: 1 kHz, -1 dBFS S8 to ASRC to ADAU1966

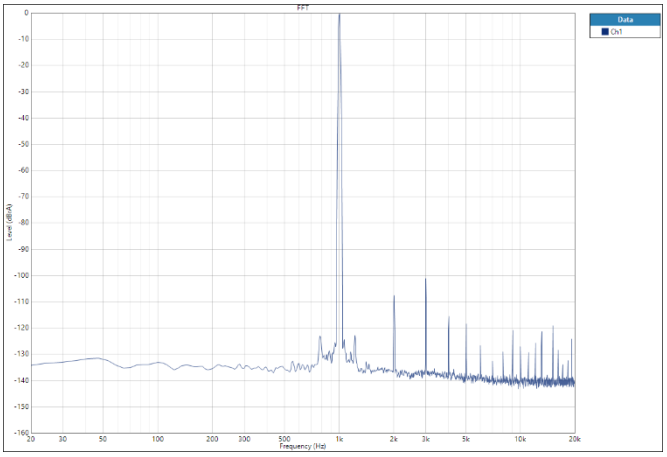


Figure 50: 1 kHz, -1 dBFS S12 to ASRC to ADAU1966

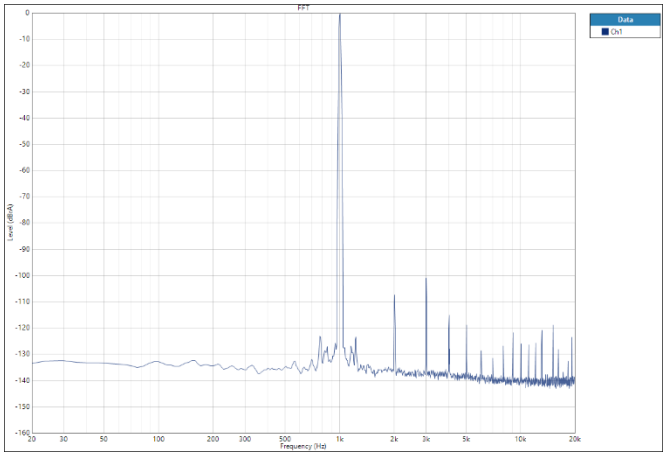


Figure 51: 1 kHz, -1 dBFS S15 to ASRC to ADAU1966

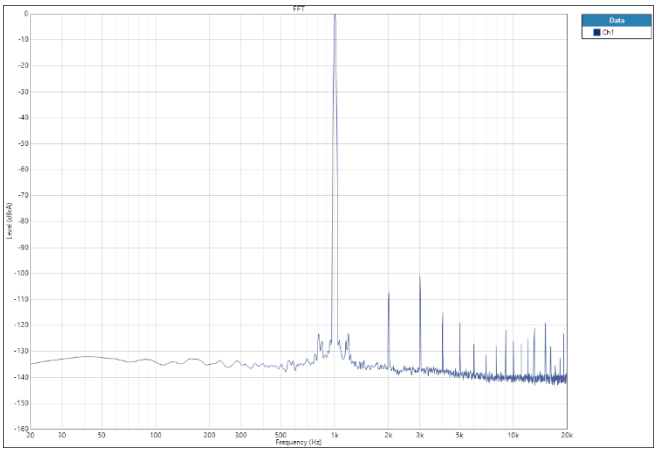


Figure 46 through Figure 51 Observations. The resampling of the audio data to a local I²S master clock has improved the noise floor significantly for all sub nodes, matching characteristics of the spectrum of the direct I²S connection of the audio source to the DAC. The effects of removing the jitter from the A²B bus recovered clocks with the ASRC connected between the sub node and the DAC achieve favorable results.

ADAU1966 -40 dBFS Test Results with ADAU1952 ASRC0 Added Before DAC

Figure 52 through Figure 57 are the supporting FFT spectrum analysis plots for the direct I²S connection of the ADAU1966 DAC from the DSIO output of the Audio Precision APx555, plus the sub node zero, sub node four, sub node eight, sub node twelve, and sub node fifteen A²B bus configurations, with an 11 kHz input sine wave at -1 dBFS:

Figure 52: 11 kHz, -1 dBFS AP I²S out to ADAU1966

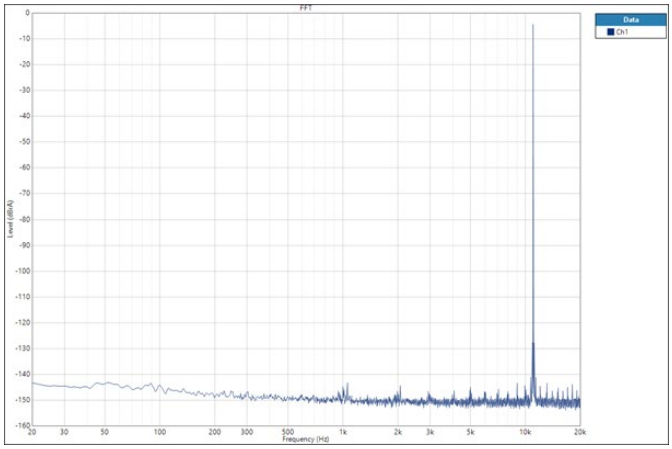


Figure 53: 11 kHz, -1 dBFS S0 I²S to ADAU1966

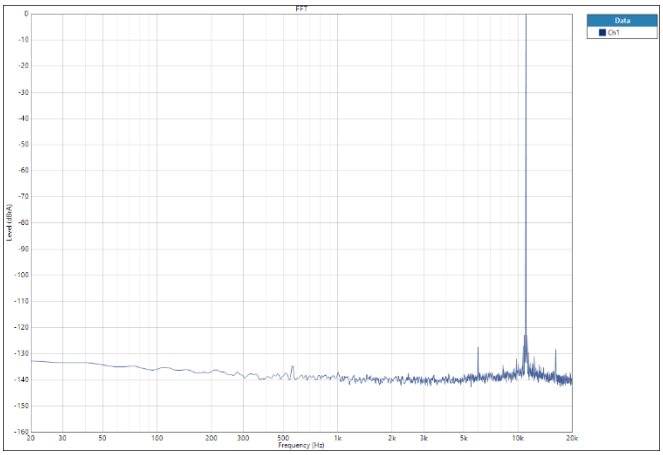


Figure 54: 11 kHz, -1 dBFS S4 I²S to ADAU1966

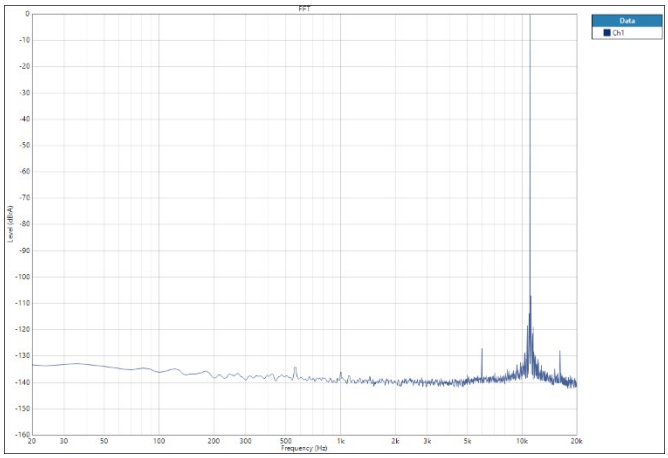


Figure 55: 1 kHz, -1 dBFS S8 I²S to ADAU1966

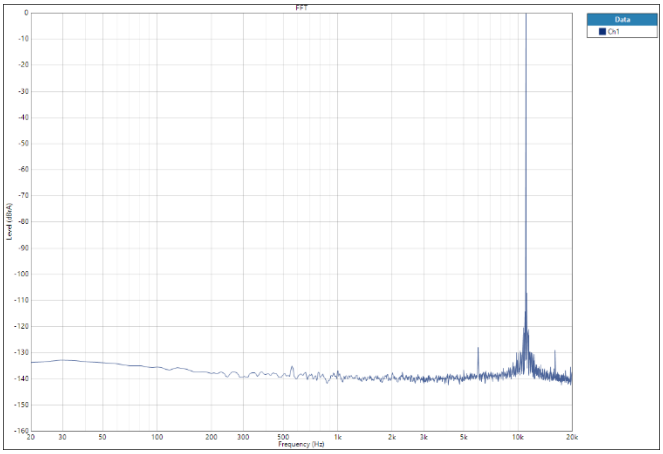


Figure 56: 11 kHz, -1 dBFS S12 I²S to ADAU1966

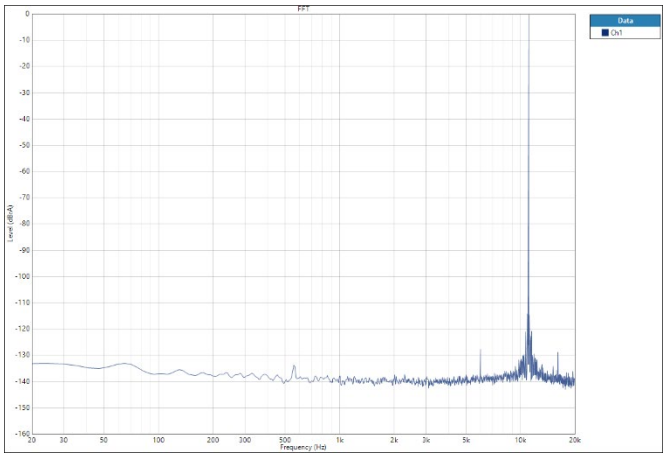


Figure 57: 11 kHz, -1 dBFS S15 I²S to ADAU1966

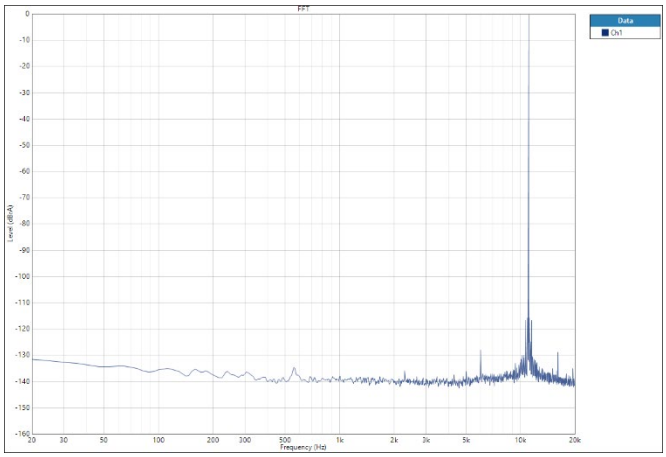


Figure 52 through Figure 57 Observations. The resampling of the audio data to a local I²S master clock has improved the noise floor and THD+N significantly for all sub nodes. Spurious noise centered close to the fundamental harmonic was observed. This noise was caused by the sample rate converter algorithm, but these do not exceed -140 dB (which matches the older AD1890/1 product specifications, so this is expected). These artifacts are unperceived by the listener and are at -120 dB or less levels. For higher frequencies, the effects of removing the jitter from the A²B bus recovered clocks with the ASRC connected between the sub node and the DAC achieve favorable results (an improved THD+N).

Analog Devices offered a discrete hardware asynchronous sample rate converter solution back in the 1990s with products such as the AD1890 and AD1891. (See *AD1890/AD1891 SamplePort Stereo Asynchronous Sample Rate Converters Datasheet*[14] for more details.) These ASRC hardware blocks have since been integrated in many larger ADI audio processors. For example, the SHARC® processors and SigmaDSP® audio processors integrate multiple hardware ASRCs that allow rate conversion of multiple A²B streams for a given A²B sub node. The ADAU1452 SigmaDSP offers a convenient option to evaluate hardware sample rate conversion that can be used to test jitter reduction and improvements on THN+N performance. The ADAU1452 device is already included in some of the local-powered sub node A²B evaluation kits provided by Analog Devices. The setup and programming of the ADAU1452 can be accomplished using the same SigmaStudio+ software tools as the AD2437.

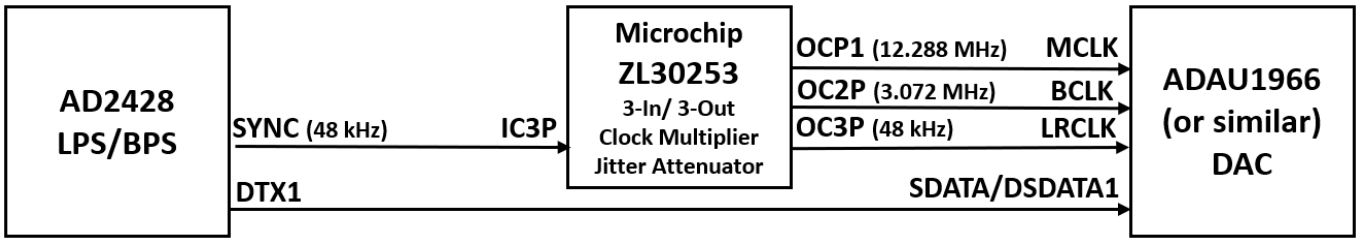
SOLUTION #2 – Using a Jitter Attenuator with a Low-Jitter Fractional PLL on A²B I²S Clocks

Another option to remove the jitter from the recovered A²B clocks is to use a jitter attenuator (JA) containing a low-jitter, low-bandwidth phase-locked loop. Combined clock multiplier and jitter attenuation devices such as the Microchip ZL30253 (formerly Microsemi, see *ZL30252, ZL30253 3-Input, 3-Output Any-to-Any Clock Multiplier and Jitter Attenuator ICs Datasheet*[6]) are great options to eliminate accumulated-node jitter from an A²B-recovered I²S clock sources. This device can take a maximum of three clock inputs and generate output frequencies for three outputs. There are other options in the ZL3025x family that can also be considered.

The block diagram in Figure 58 shows an example system configuration where the 48 kHz SYNC (I²S left/right clock) feeds into the JA and uses the clock multipliers to generate the necessary I²S output clocks with less than 0.3 picoseconds of RMS jitter. For 24-bit stereo I²S data received from an A²B sub mode, the

ZL30253 needs to generate a 12.288 MHz MCLK, a 3.072 MHz BCLK, and a 48 kHz LRCLK that is required to clock an audio DAC, such as the ADAU1966. The output 48 kHz I^2S left/right clock can be phase aligned to the 48 kHz SYNC from the AD2437. This also would guarantee phase alignment of the 3.072 MHz BCLK and 12.288 MHz MCLK as they are integer multiples of 48 kHz.

Figure 58. Block Diagram for Connecting the AD2437 to a DAC Using a ZL30253 JA to Remove SYNC Jitter



Another benefit of this family of JAs is a holdover feature where it continues to generate the output clocks when there is a loss of the input clock. This added feature is useful in an A^2B system when there is a system reset or rediscovery of the A^2B bus. The ZL3025x JA can still maintain the I^2S clocks to the DACs and prevent issues like audible clicks or pops on output speakers, and the DAC outputs can be muted until the A^2B bus is rediscovered.

The benefits of the ZL3025x JA device are:

- Offer jitter reduction of the A^2B -recovered I^2S clocks.
- Hold over and maintain the input reference clock when the A^2B main node clock is temporarily disabled, removing audible glitches in the system. Contact Microchip for more information regarding other lower-cost variants of this JA/clock multiplier family.

Spread-Spectrum Clocking

In order to meet the demanding automotive standards for EMC, system designers must take care to reduce emissions caused, in part, by low rise-time digital signals wherever possible. A simple method for mitigating electromagnetic radiation problems in digital clocks is to use frequency modulation of the clock signal to smear the power across a wider bandwidth, reducing the peak power at any one frequency and thus reducing radiated emissions. This mitigation technique is called Spread-Spectrum Clocking.

The AD242x family of products provides system designers several key parameters that can be tuned to implement this spread-spectrum clocking in order to improve EMC performance. The mode parameter for spread spectrum allows for three options for selecting which signals will be modulated. The default mode setting is for no spread spectrum to be applied. Modes 2 and 3 give the user the option to enable spread-spectrum clocking only for the clocks related to the A^2B bus itself as well as the option to apply it to both A^2B clocks as well as the external clocks for the I^2S/TDM port, respectively. Additionally, there are two parameters for adjusting the modulation that is used, the depth and the frequency of the modulating signal. A designer is able to modulate clocks using two depth options, low and high, as well as four different frequency options, 4x, 5x, 6x, and 7x. These frequency options are integer multiples of the SYNC signal which sets the period of the data superframe.

Effects of Spread Spectrum Clocking

By modulating the clock frequency slightly, emissions can be reduced but this option also introduces another possible source of jitter. To evaluate the impacts of spread spectrum, testing similar to that which was done in previous sections was performed while also sweeping the spread spectrum settings. As seen in Figure 59, at high signal levels, these settings appear to have minimal impact on the audio performance of the overall A^2B system.

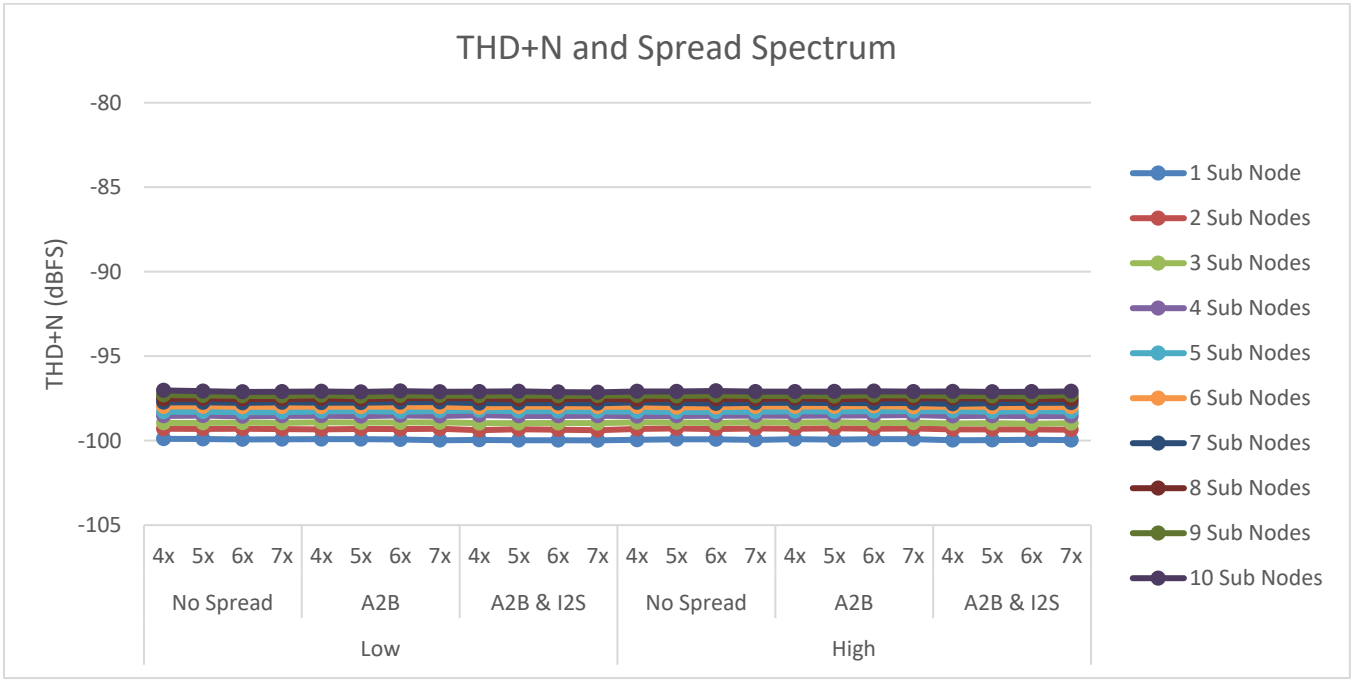


Figure 59. Effects of Spread Spectrum at -1 dBFS

Additionally, at low signal levels, -40 dBFS, we also see minimal impact from enabling spread spectrum. THD+N holds steadily around -114 dBFS for all cases. This is shown in Figure 60 below.

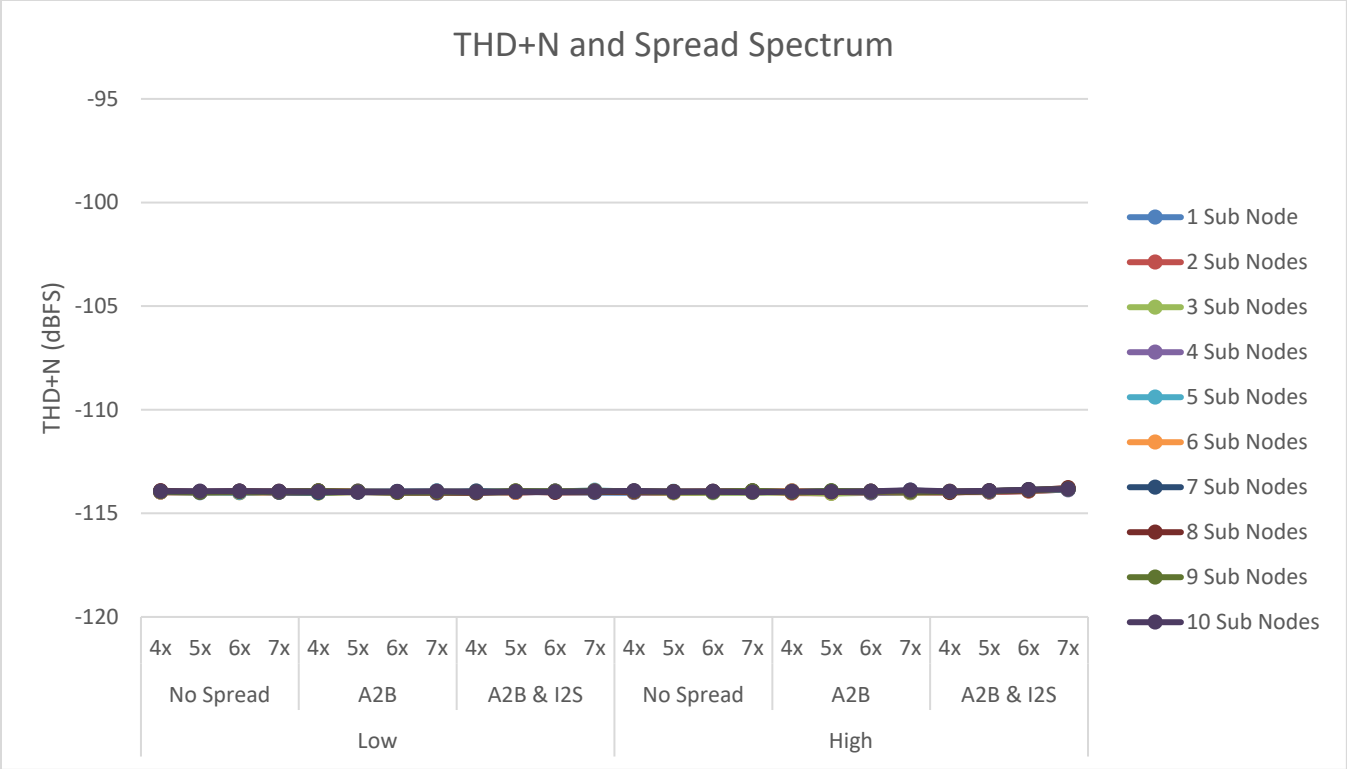


Figure 60. Effects of Spread Spectrum at -40 dBFS

[The author would like to acknowledge and thank *Michael Hathaway* for his contribution to the Spread Spectrum performance testing and results used in this section of this EE-Note]

Conclusion

In A^2B systems, node-to-node jitter is unavoidable. These audio-networked systems accumulate jitter at each subordinate node, but typical amounts of such error are generally lower than those incurred by a DAC or ADC. Because overall system audio performance appears to be dominated by the DAC performance, THD+N measurements can be significantly influenced by the harmonic distortion introduced by the DAC at higher signal levels with the degradation quickly falling off as the amplitude is reduced. Thus, higher levels of clock jitter can begin to have an impact on the performance of the analog audio output of the system. However, the resulting degradation effect on audio performance seen at the downstream A^2B bus nodes can be reduced by selecting a converter with a low-bandwidth PLL to remove the upper frequency components of the jitter on the recovered clock. Jitter Attenuator devices are another option to remove the jitter from the recovered clocks and prevent signal degradation. Audio degradation can even be further reduced or eliminated with processors containing HW ASRC blocks, since the resampled audio data is instead driven from clocks derived from a clean local clock oscillator source. Finally, at measured levels of THD+N seen here, spread-spectrum clocking appears to show minimal audio degradation regardless of modulation parameters with performance showing only very minor variation when sweeping spread-spectrum settings, even at high node counts. Thus, spread-spectrum modes on the AD2428 have a negligible impact on an audio converter's THD+N performance.

References

[1] *AD2426/AD2427/AD2428 A²B Transceiver Datasheet (Revision F), 2023. Analog Devices, Inc.* https://www.analog.com/media/en/technical-documentation/data-sheets/ad2426-ad2427-ad2428.pdf

[2] *AD2426/7/8 Automotive Audio Bus (A2B) Transceiver Technical Reference (Rev.1.3), June 2023. Analog Devices Inc.* https://www.analog.com/media/en/technical-documentation/user-guides/ad242x-trm.pdf

[3] *EVAL-AD2428WD1BZ A 2B Evaluation Board Manual, Revision 1.1, October 2019. Analog Devices Inc.* https://www.analog.com/media/en/technical-documentation/user-guides/EVAL-AD2428WD1BZ_Manual.pdf

[4] *ZL30252, ZL30253 3-Input, 3-Output Any-to-Any Clock Multiplier and Jitter Attenuator ICs Datasheet, January 2018, Microsemi.* https://ww1.microchip.com/downloads/en/DeviceDoc/ZL30252,_ZL30253_data_sheet_012318.pdf

[5] *ADAU1966 16-Channel High Performance Differential Output, 192 kHz, 24-Bit DAC Datasheet, Revision E, March 2016. Analog Devices, Inc.* https://www.analog.com/media/en/technical-documentation/data-sheets/ADAU1966.pdf

[6] *Evaluating the ADAU1962/ADAU1966 High Performance, Low Power, Multibit Sigma-Delta DACs, Evaluation Board User Guide/UG-416. Revision A, January 2014. Analog Devices, Inc.* https://www.analog.com/media/en/technical-documentation/user-guides/UG-416.pdf

Further Reading

[7] *Technote 008-A²B I²S Clock Jitter, Revision 2, July 2020. Clockworks Signal Processing, LLC.* https://clk.works/wp-content/uploads/TechNotes/TN008_A2B_jitter.r2.pdf

[8] *Technote 010-AK4619 ADC and DAC Performance - Evaluation of the Impact of A²B I²S Clock Jitter, Revision 2. May 2023. Clockworks Signal Processing LLC.* https://clk.works/wp-content/uploads/TechNotes/TN010_A2B_ak4619_performance.r2.pdf

[9] *Appnote 001 - Measuring Jitter Spectrum with a DSO, Revision 1.0b, July 2020. Clockworks Signal Processing LLC.* https://clk.works/wp-content/uploads/AppNotes/AP001jitter_measurement.r1b.pdf

[10] *N. Rocchi, A. Toscani, G. Chiorboli, D. Pinardi, M. Binelli, and A. Farina, "Transducer Arrays Over A²B Networks in Industrial and Automotive Applications: Clock Propagation Measurements," in IEEE Access, vol. 9, pp. 118232-118241, 2021, doi: 10.1109/ACCESS.2021.3106710. Unverified link.* https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9520394

[11] *AD1890/AD1891 SamplePort Stereo Asynchronous Sample Rate Converters Datasheet, Revision 0, 1993, Analog Devices, Inc.* https://www.analog.com/media/en/technical-documentation/data-sheets/AD1890_1891.pdf

[12] *Robert Adams, Jitter Analysis of Asynchronous Sample-rate Conversion, Presented at 95th AES Convention, Oct 7-10, 1993. (Link to Purchase)* https://www.aes.org/e-lib/browse.cfm?elib=19086

[13] *Adams, Robert & Kwan, Tom, Theory and VLSI Architectures for Asynchronous Sample-Rate Converters, Journal of the Audio Engineering Society. JAES Volume 41 Issue 7/8 pp. 539-555; July 1993. (Link to Purchase)* https://secure.aes.org/forum/pubs/journal/?elib=6993

[14] *Kevin James McLaughlin and Robert Adams, An Asynchronous Sample Rate Converter with 120 dB THD+N Supporting Sample Rates up to 192 kHz. Analog Devices, Wilmington, MA, USA. Presented at the 109th AES Convention, September 2000. (Link to Purchase)* https://www.aes.org/e-lib/browse.cfm?elib=9147

[15] *Maury Wood, Analysis of Jitter Rejection of SRCs and DACs Using an NCO, Analog Devices Inc., Wilmington, USA, Presented at the 100th AES Convention, May 1996. (Link to Purchase)* https://www.aes.org/e-lib/browse.cfm?elib=7515

[16] *Boris Lerner and Aaron Lowenberger, EE-261: Understanding Jitter Requirements of PLL-Based Processors, Revision 1, January 2005, Analog Devices, Inc.* https://www.analog.com/media/en/technical-documentation/application-notes/EE-261.pdf

[17] *Chris Travis, Paul Lesso, Specifying the Jitter Performance of Audio Components, Presented at the 117th Convention, October 2004, USA Audio Engineering Society.* https://www.aes.org/tmpFiles/elib/20231107/12950.pdf

[18] *Dunn, Julian, Jitter and Digital Audio Performance Measurements. UK 9th Conference: Managing the Bit Budget, May 1994, Audio Engineering Society,* https://www.aes.org/e-lib/browse.cfm?elib=6111

[19] *Julian Dunn, Jitter Theory-Technote 23. Audio Precision, Inc. AudioPhilleo, LLC. Knowledge Base.* http://www.audiophilleo.com/zh_hk/docs/Dunn-AP-tn23.pdf

[20] *James A. S. Angus, A New Method for Analyzing the Effects of Jitter in Digital, Audio Systems, 103rd AES Convention, May 1998, Audio Engineering Society.* https://www.aes.org/e-lib/browse.cfm?elib=8464

[21] *Eric Benjamin and Benjamin Gannon, Theoretical and Audible Effects of Jitter on Digital Audio, Quality, Dolby Laboratories Inc. San Francisco, CA 94103, USA, Presented at the 105th AES Convention, September 1998, Audio Engineering Society.* https://www.aes.org/e-lib/browse.cfm?elib=8354

[22] *Bruno Putzeys and Renaud de Saint Moulin. Effects of Jitter on AD/DA Conversion. Specification of Clock Jitter Performance. Presented at the 116th Convention, May 2004, Audio Engineering Society. (Link to Purchase)* https://secure.aes.org/forum/pubs/conventions/?elib=12637

[23] *EVAL-ADAU1452 Obsolete Evaluation Board, Analog Devices, Inc.* https://www.analog.com/en/design-center/evaluation-hardware-and-software/evaluation-boards-kits/eval-adau1452.html

Document History

Revision	Description
Revision 1 – September 20, 2024 *by John Tomarakos*	Initial Release